

MBA Thesis

A Comparative Analysis of AI and Dot-com Eras

Involving Market Indices and Financial Fundamentals of Firms

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The author(s) declare(s) that they have completed the thesis work independently. All external sources are cited and listed under the References section. The thesis work has not been submitted in the same or similar form to any other institution(s) as part of another examination or degree.

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Abstract

Artificial Intelligence (AI) has attracted strong investor attention and substantial capital allocation. This raises a practical question familiar from earlier technology cycles: whether current market pricing for AI mainly reflects justified growth expectations or also contains speculative elements comparable to the dot-com era. The thesis addresses this issue by comparing market-level index behavior and firm-level valuation and profitability patterns across the dot-com period and the current AI cycle.

Background: Technology-driven optimism can produce periods in which equity prices move ahead of visible fundamentals. The dot-com bubble is a canonical benchmark for such a cycle. The AI boom may share some features with that episode, but it may also differ due to firms' maturity, profitability, and broader adoption of the technology across sectors.

Objectives: The purpose of this study is to evaluate, using historical and descriptive evidence, to what extent the current AI hype resembles the dot-com bubble in terms of (i) market-level cycle behavior in major U.S. equity indices and (ii) firm-level valuation multiples and operating profitability for AI-positioned firms relative to non-AI controls and dot-com benchmark groups.

Methodology: A comparative quantitative design combines two perspectives. At the market level, the study compares the National Association of Securities Dealers Automated Quotations (NASDAQ) Composite, NASDAQ 100, and Standard Poor's (SP) 500 across the dot-com and AI periods using normalized index paths; correlation and dynamic time warping distance on LOcally Weighted Scatterplot Smoothing (LOWESS)-smoothed, standardized logarithmic trends; rolling dynamic time warping similarity between the AI-era path and segment-wise dot-com cycle phases; and joint analysis of rolling volatility and log returns with kernel density visualization. At the firm level, it compares Price to Sales Ratio (P/S), Price to Earning (P/E), enterprise value to Enterprise Value to EBITDA (EV/EBITDA), and Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA) margin across defined groups: dot-com bubble versus dot-com control firms, and AI-positioned versus non-AI control firms for the AI era, with group rules based on archival dot-com lists and external AI-positioning scores (Larridin AI Tracker, with complementary context from adoption indices). Most firm-level multiples and related fields were assembled from historical records in Macrotrends, supplemented by the external trackers used to classify AI positioning; the analysis is descriptive and does not estimate intrinsic value or causal effects.

Results: Market-level evidence suggests partial similarity between the AI period and specific phases of the dot-com cycle (notably acceleration in some windows), but overall trend similarity is moderate (correlation on smoothed trends roughly 0.59–0.63) rather than an exact replay. The AI-era index path appears comparatively gradual, with episodic corrections, while the dot-com episode shows a more complete boom–bust profile in the analyzed technology-heavy indices; volatility–return patterns share a central region of low-to-moderate volatility near zero returns but differ in tail dispersion. Firm-level multiples indicate that AI-positioned firms are not valued at the extremes observed for dot-com bubble firms; gaps between AI-positioned and non-AI controls are limited, especially when valuation is related to operating earnings. EBITDA-based profitability indicators point to stronger operating performance in the AI-era sample than in many dot-com bubble firms. Recovery analysis for the dot-com crash further suggests that less extreme valuations were associated with faster post-correction recovery, though the AI cycle is still ongoing.

Conclusions: Taken together, the evidence supports a nuanced answer to the main research question: the AI period shows some bubble-like signals and limited market-level resemblance to parts of the dot-com cycle, but it should not be characterised as a straightforward repetition of the dot-com bubble based on the indicators used. The balance of index and firm-level results is more consistent with a technology boom in which expectations are partly priced, yet are often linked to more mature firms and stronger reported operating fundamentals than in the dot-com peak. Interpretation remains subject to survivorship bias in the historical sample, incomplete outcome data for an ongoing AI cycle, and the use of external proxies for AI positioning.

Recommendations for future research: Future work could extend the comparison to other historical episodes, refine AI-exposure measurement, improve coverage of delisted dot-com firms where data allow, incorporate private-market AI valuations, and revisit market- and firm-level patterns as post-peak outcomes for the AI cycle become observable.

Keywords: AI, dot-com bubble, market index, valuation multiples, comparative analysis, speculation

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Contents

Abstract i

Acknowledgements ii

List of Abbreviations v

List of Symbols vii

List of Tables viii

List of Figures ix

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 Problem Discussion . . . 2
- 1.3 Research Purpose . . . 3
- 1.4 Scope and Delimitations . . . 4
- 1.5 Thesis structure . . . 4

2 Prior Work and Theoretical Framework 5
- 2.1 Technology-Driven Bubbles and Market Cycles . . . 5
  - 2.1.1 Overvaluation and the Anatomy of a Bubble . . . 5
  - 2.1.2 Technological Diffusion and the Diffusion-Evaluation Gap . . . 5
  - 2.1.3 Market Cycles and Bubble Phases . . . 5
- 2.2 AI-Positioned Firms and Valuation Uncertainty . . . 6
  - 2.2.1 AI as a General Purpose Technology . . . 6
  - 2.2.2 Measuring AI-Positioning . . . 6
  - 2.2.3 Valuing AI-Positioned Firms . . . 7
- 2.3 Valuation Multiples and Speculative Markers . . . 7
  - 2.3.1 Valuation Challenges and Financial Multiples . . . 7
  - 2.3.2 P/S, P/E, EV/EBITDA, and EBITDA Margin . . . 8
  - 2.3.3 Investor Sentiment and Valuation Plausibility . . . 8
- 2.4 Market Cycles Using Similarity Measures of Indices . . . 9
- 2.5 Research Gap . . . 10
- 2.6 Synthesis and Hypothesis Development . . . 11

3 Methodology 12
- 3.1 Research Design . . . 12
- 3.2 Data Collection . . . 12
  - 3.2.1 Market-Level Data Collection . . . 12
  - 3.2.2 Firm-Level Data Collection . . . 13
- 3.3 Variable Construction and Measures . . . 14
  - 3.3.1 Group Definitions . . . 14
  - 3.3.2 Financial Multiples and Indicators . . . 14
  - 3.3.3 Descriptive and Comparative Indicators . . . 15
- 3.4 Analysis Procedure . . . 15
  - 3.4.1 Market-Level Analysis Procedure . . . 15
  - 3.4.2 Firm-Level Analysis Procedure . . . 17
- 3.5 Reliability, Validity, and Limitations of the Collected Data . . . 17
- 3.6 Ethical Considerations . . . 18
- 3.7 Summary . . . 18

4 Results and Analysis **19**
4.1 Historical Market-Level Analysis: Indices Comparison . . . 19
4.1.1 Normalized Evolution of the AI and Dot-com Eras . . . 20
4.1.2 Similarity Measures Between the AI and Dot-com Eras . . . 21
4.1.3 Index Similarity of the AI era with the Composing Phases of the Dot-Com Bubble 22
4.1.4 Volatility and Return Comparison between the AI and Dot-com Eras . . . 24
4.2 Firm-Level Analysis: Financial Multiples Comparison . . . 26
4.2.1 Descriptive Profile of the Compared Firms . . . 26
4.2.2 Price-to-Sales Comparison . . . 27
4.2.3 Price-to-Earnings Comparison . . . 28
4.2.4 EV/EBITDA Comparison . . . 29
4.2.5 EBITDA Margin Comparison . . . 30
4.2.6 Recovery Perspectives . . . 30

5 Discussion **31**
5.1 Key Findings . . . 31
5.2 Evidence Against a Direct Dot-Com Bubble Repetition . . . 32
5.2.1 Stronger Fundamentals in the AI-Positioned Firms . . . 32
5.2.2 Broader AI Adoption Across Sectors . . . 33
5.2.3 Limited Separation Between AI-Positioned Firms and Controls . . . 33
5.2.4 A More Sustainable Market Evolution . . . 34
5.3 Bubble Signals in the AI Era . . . 34
5.3.1 Market Structure, Breadth, and Volatility . . . 34
5.3.2 Investor Expectations Around AI . . . 35
5.4 Interpretation in Relation to the Research Questions . . . 35
5.5 Findings in Relation to Previous Research . . . 36
5.6 Reflections and Limitations . . . 38
5.6.1 Market-Level Methodological Limitations . . . 38
5.6.2 Firm-Level Data and Sampling Limitations . . . 38

6 Conclusion and Future Research **39**
6.1 Summary . . . 39
6.2 Conclusion . . . 39
6.3 Research Contributions . . . 40
6.4 Future Work . . . 40

References **41**

A Appendix: Figures **47**
A.1 Market-Level Index Figures . . . 47
A.2 Firm-Level Recovery . . . 49
A.3 Firm-Level Comparability Analysis . . . 49
A.3.1 Price-to-Sales Controlled for Industry and Maturity . . . 49

B Appendix: Tables **52**
B.1 Previous Research Comparison Table . . . 52
B.2 Chapter 4 Tables . . . 52

List of Abbreviations

AI Artificial Intelligence

AIMI AI Mispricing Index

ARDL stands for Autoregressive Distributed Lag

AR-X Autoregressive model with eXogenous variables

CapEx Capital Expenditure

DCF Discounted Cash Flow

DSGE Dynamic Stochastic General Equilibrium

DTW Dynamic Time Warping

EBITDA Earnings Before Interest, Taxes, Depreciation, and Amortization

ETF Exchange Traded Fund

EV Enterprise Value

EV/EBITDA Enterprise Value to EBITDA

EVR Enterprise Value to Revenue

FCF Free Cash Flow

FuV Fundamental Value

GARCH Generalized Autoregressive Conditional Heteroskedasticity

GCN Graph Convolutional Network

GPTech General Purpose Technology

GSADF Generalized Supremum Augmented Dickey-Fuller

IP Intellectual Property

IQR Inter-Quartile Range

KDE Kernel Density Estimation

LOWESS LOcally Weighted Scatterplot Smoothing

LSTM Long Short-Term Memory

MV Market Value

NASDAQ National Association of Securities Dealers Automated Quotations

NYSE New York Stock Exchange

P/E Price to Earning

PEG Price/Earnings-to-Growth

P/S Price to Sales Ratio

PV Present Value

RD Research and Development

ROV Real Option Valuation

RQ Research Question

SMA Simple Moving Average

SNA System of National Accounts

SP Standard Poor's

TMT Technology, Media, and Telecommunications

VIX Volatility IndeX

List of Symbols

r	Discount rate
r_i	Returns of the i-th day
σ_i	Volatility of the i-th day
$\sigma_{\text{annualized},i}$	Annualized volatility of the i-th day
$\mathbf{x}$	Time series or a vector
$\mathbf{H}$	Matrix
x_i	Discrete samples of a time series
z_i	Standardized discrete samples of a time series
P_i	Close price of the i-th day
σ_i	Standard diviation
ΔQ	The difference between the third quartile and the first quartile
$r_{\mathbf{x},\mathbf{y}}$	Correlation between the time series $\mathbf{x}$ and $\mathbf{y}$
$\hat{f}(\cdot,\cdot;\cdot)$	Kernel Density Estimation
$K_{\mathbf{H}}(\cdot)$	Bi-variate kernel function

List of Tables

2.1 AI-Positioning Assessment Dimensions 7
2.2 Valuation approaches for AI-positioned firms 7

4.1 Similarity Measures Between Dot-com and AI Era Market LOWESS Standardized Trends 22
4.2 Sample size and founding-year profile of compared firm groups 26
4.3 Largest sector categories within each compared firm group 27
4.4 Mean and median P/S ratios across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms) 27
4.5 Mean and median P/E ratios across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms). 28
4.6 Recovery years across dot-com bubble and dot-com control firms 30

B.1 Summary of previous Literature on AI vs. Dot-com Bubbles 52
B.2 Mean and median EV/EBITDA multiples across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms) 53
B.3 Mean and median EBITDA margins across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms) 53

List of Figures

1.1 Aggregate private investment in AI in the United States, Europe, and China, 2013–2028, with data harvested from (Stanford Institute for Human-Centered Artificial Intelligence, 2026). Note that the values for 2026–2028 are illustrative linear predictions based on the 2013–2025 trend. 2

4.1 Normalized Index Performance: Dot-com vs AI Era 21
4.2 Rolling DTW distance between the AI era (2016 to 2026) and the segments of the dot-com era that correspond the the phase of early adoption (1995–1997), acceleration (1998), mania (1999–March 2000), crash (March 2000–2002) and consolidation (2003–2005) for the indices of NASDAQ Composite, NASDAQ 100, and SP 500. 23
4.3 Returns and rolling volatility (annualized) for the indices NASDAQ Composite, NASDAQ 100, and SP 500 using scatters, the Kernel Density Estimation (KDE) of the dot-com era, and the KDE of the AI era. 25
4.4 Distribution of P/S ratios across dot-com and AI firm groups. 28
4.5 Distribution of P/E ratios across dot-com and AI firm groups. 29
4.6 Distribution of EV/EBITDA multiples across dot-com and AI firm groups. 29
4.7 Distribution of EBITDA margins across dot-com and AI firm groups. 30

A.1 NASDAQ Composite (Dot-com Era): Close, Simple Moving Average (SMA)(50), LOWESS , and volume 47
A.2 NASDAQ Composite (AI Era): Close, SMA(50), LOWESS , and volume. 47
A.3 NASDAQ 100 (Dot-com Era): Close, SMA(50), LOWESS, and volume. Note that, until the moment of writing, the volume data for NASSDAQ 100, TradingView (nd) was available since the second part of 1999. 48
A.4 NASDAQ 100 (AI Era): Close, SMA(50), LOWESS , and volume. 48
A.5 SP 500 (AI dot-com): Close, SMA(50), LOWESS, and volume. 48
A.6 SP 500 (AI Era): Close, SMA(50), LOWESS , and volume. 48
A.7 Distribution of the years that took firms' stock prices to recover after the dot-com bubble. 49
A.8 Industry Composition by Group 49
A.9 Median P/S by group for tech firms only 50
A.10 Median P/S grouped by firm maturity 50
A.11 Median P/S grouped by firm maturity and valuation years 51

1. Introduction

1.1 Background

In recent years, Artificial Intelligence (AI) has gained and is still gaining increasing attention both from academia, industry, and investment community (Chrzanowska et al., 2024; Goldburgh et al., 2024) but also from policymakers. The creating strength behind this phenomenon is the potential of the AI for reshaping business, value creation (Perifanis and Kitsios, 2023; Rabhi et al., 2025), and human society in general (International Monetary Fund, 2024; United Nations Development Programme, 2025). AI is not considered anymore as a research field or a collection of software tools but rather as a General Purpose Technology (GPTech) that has the strength to increase productivity, improve over time, create complementary innovations, and cause systemic economic change (Bresnahan and Trajtenberg, 1995), but also cause societal changes (Brynjolfsson and McAfee, 2014). In terms of policymaking, several challenges that are connected with fairness, transparency, privacy, safety, and accountability are created (OECD AI Policy Observatory, 2019). According to OECD AI Policy Observatory (2019), these challenges should be resolved via regulation, investment, data governance, skills development, and international cooperation towards trustworthy, human-centered AI.

As a consequence, the previous observations have been reflecting and influencing the ongoing and previous investment activity in private and public markets. Especially, in terms of private investments, based on the Stanford Institute for Human-Centered Artificial Intelligence (2026), AI investment in the United States, Europe, and China reached $319.2 billion in 2025, compared with $137.0 billion in 2024. In more detail, as shown in Fig. 1.1, the AI investments in the United States, Europe, and China follow an increasing trend since 2013, with an exception between years 2021 and 2022. Using a linear prediction, the same investments are expected to reach the level of $521.2 billion in 2027 and $633.8 billion in 2028. Regarding adoption, the Stanford Institute for Human-Centered Artificial Intelligence (2026) states that organizational AI adoption has already reached the level of 88%, while the generative AI has reached a level of population adoption of 53% within three years . These numbers show that AI has already become both a technological and financial phenomenon, while its impact is increasing as time evolves. As investors are not only reacting to present earnings, but also to expected future growth and productivity gains (Campbell and Shiller, 1988), the valuation of investments and future returns is a challenging task which includes high levels of uncertainty (Dixit and Pindyck, 1994).

From the above discussion, it is clear that AI is more than a simple technology that attracts temporal attention. More specifically, AI is framed as a GPTech (Brynjolfsson et al., 2021) with extensive transformation power that goes beyond productivity gains and firm value creation (Brynjolfsson and McAfee, 2014). Since 2013, the increasing investments, shown and predicted in Fig. 1.1, are justified by the accelerating adoption of AI into the operational processes, product development, customer interaction, data analytics, automation, and strategic decision-making of firms (Enholm et al., 2022; Machucho and Ortiz, 2025). Therefore, the finding of this thesis may also provide arguments that set AI as a (digital) transformational technology that can set a new norm on how business, science, etc, is be done.

With respect to valuation and the impact that AI has on the valuation, there are already contributions that claim connections between AI and value (Koulis et al., 2025). Thus, a complete valuation needs to consider also the level of AI exposure of a firm and allocate the appropriate AI premium. Clearly, such a task is challenging both in terms of measuring the AI positioning but also in terms of pricing the AI premiums. According to (Babina et al., 2024), the AI positioning of a firm is connected with its public disclosures that are relevant to AI activities and the use of capital for AI activities. This indicates that AI positioning is visible through firm behavior, public communication, and investment in AI capabilities. At the same time, AI, like other emerging technologies, poses additional valuation challenges due to future uncertain benefits which might require extensive time to materialize (Fernandez, 2023; Kaartemo and Nyström, 2021).

In general, technology-driven optimism has historically created periods in which market expectations move faster than the financial fundamentals of firms (Perez, 2002). More specifically, Lansing (2008) argues that investors often overreact to technological innovations and exaggerate the improvement in economic fundamentals. Such a reaction may lead to the formation of bubbles. The dot-com bubble is a

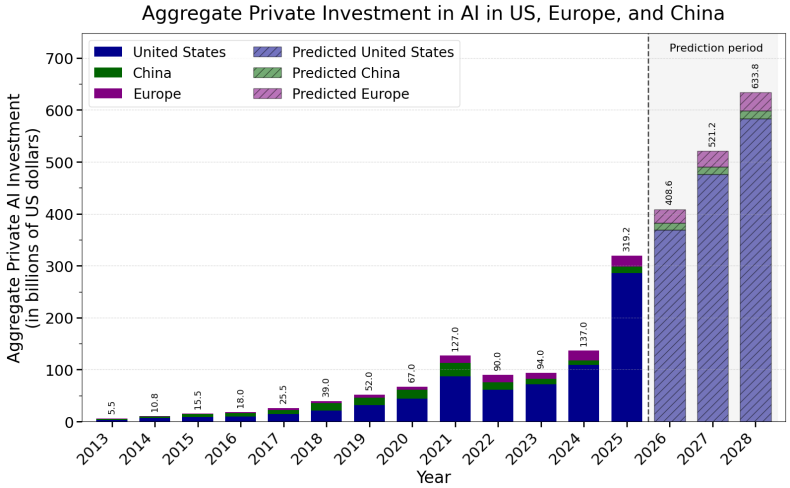


Figure 1.1: Aggregate private investment in AI in the United States, Europe, and China, 2013–2028, with data harvested from (Stanford Institute for Human-Centered Artificial Intelligence, 2026). Note that the values for 2026–2028 are illustrative linear predictions based on the 2013–2025 trend.

central example of such a period, where expectations about the Internet and digital transformation led to strong market enthusiasm, high valuations, and later market corrections. During this period, stocks traded at earnings multiples well above historical levels, weakening the relationship between market value and traditional accounting information (Morris and Alam, 2012).

The previous historical and research evidence highlights the importance of exploring whether the current AI hype is a typical example that can lead to the creation of a bubble. In addition, if the market is in the process of bubble formation, an important question to answer is the proximity of the market to this point. One way to answer the previous question is to identify whether valuations today are supported by stronger fundamentals or whether they partly reflect speculative expectations. Additional conclusions can be drawn by studying the behavior of the most representative indices, either in terms of their actual characteristics or in terms of the characteristics of another benchmark that is a known bubble. Such an example is the dot-com era. Motivated by this, the current thesis aims to answer the previous questions using quantitative arguments, which are also supported by well-established theoretical frameworks.

1.2 Problem Discussion

Historically, the market has shown that technology-driven hype, such as the dot-com bubble, can create a market cycle where technology firms are overpriced, followed by a subsequent market correction (Maghyereh and Awartani, 2025; Shiller, 2015). As discussed above, due to uncertainty, the main challenge is to distinguish between justified optimism and speculative overvaluation while a technology cycle is still under development. Despite the potential for real economic value, financial markets may also price in these expectations too early or too aggressively (Pastor and Veronesi, 2009).

Such a situation is especially relevant for AI. On one hand, AI is already showing signs of improved productivity, reduced costs, and creation of new products, along with a change in strategy. On the other hand, there is high uncertainty with respect to the speed of financial evolution of AI, especially due to high capex (Caballero, 2026). In (Albori et al., 2025), it is argued that the current U.S. technology sector should be assessed by asking whether the growth required to justify current prices is reasonable compared to historical experience. This is important because high prices may be justified if future earnings growth is strong enough and at a pace higher than the increase of capex, but they may also indicate overvaluation if expectations become unrealistic or if the rate of increase of capex is not attainable by growth.

The experience from the dot-com era clearly shows that the previous thoughts matter. During the late 1990s, technology stocks increased sharply. As shown by (Griffin et al., 2011), the institutional investors, despite their assumed financial intellect, played an important role in driving technology stock prices during the boom. Later, many firms failed to deliver the expected fundamentals, and market values corrected with a bubble burst. This does not mean that the Internet was unimportant, but that

a transformation technology can be real while parts of the market valuation around it are still excessive.

Today, there is a danger that the AI hype may follow some similar patterns, where the long-term valuation of AI-positioned firms may reflect both real growth potential and speculation. The uncertainty is increased because the economic impact of AI is still developing and because the current AI cycle has not yet completed a full acceleration, peak, correction, and recovery path. Therefore, it is too simple to ask only whether AI is important. The more relevant problem is whether the financial market is valuing AI-related growth in a way that resembles earlier technology-driven bubbles.

In this thesis, the terms AI hype or AI era refer to the current period in which AI receives strong attention from firms, investors, and public markets. A firm is categorized as AI positioned when it shows strong observable exposure to AI according to external AI-ranking data and/or public indicators of AI adoption. The influence of AI on the pricing of a firm can be studied both via the price evolution of the firm or via its multiples. At the market level, the influence of AI can be studied via the evaluation of related indices. For both cases, among others, signals of a bubble creation can be identified by using historical benchmarks. An excellent historical benchmark of a technology-driven market cycle in which expectations were strongly reflected in market prices without being supported by fundamentals is the dot-com bubble. These concepts are connected through the central issues of: a) valuation which is whether market prices and financial multiples are supported by fundamentals or if they resemble earlier speculative patterns that may lead to bubbles and b) the evolution of indices that may lead to a bubble.

Against the previous framework, the identification of signals of speculation can be done both at the market level or firm level. At the market level, it may appear through index behavior, such as rapid increases, concentration, volatility, or similarity to earlier bubble cycles. At the firm level, it may appear through financial multiples and profitability indicators, where AI-positioned firms may be priced differently from comparable firms. Looking at only one level may give an incomplete picture. Index analysis can show broad market behavior, but it cannot explain whether individual AI-positioned firms are fundamentally stronger or simply more expensive. Therefore, there is a clear motivation for a study which, systematically and based on historical data, assesses the AI boom. By combining a market-level comparison with a firm-level comparison, this study seeks to examine whether the AI era resembles the dot-com period in broad market behavior and whether similar patterns can also be observed in the valuation of individual firms. This is important because it helps to assess whether the current AI boom is mainly supported by real technology adoption, driven by speculative expectations, or shaped by a combination of both. To the best of our knowledge, although there extensive research on the topics of technology bubbles (Griffin et al., 2011; Weitzel et al., 2020), investor sentiment (Baker and Wurgler, 2006; Maurya et al., 2026), valuation multiples (Cooper and Lambertides, 2023), and AI adoption (OECD, 2025a), separately, the academic literature related to the joint quantitative comparison of the AI era with the dot-com bubble using both index and firm-level fundamental information information is still limited.

Based on this discussion, it is clear that this study contributes to the field of industrial economics and management by connecting technology adoption, market cycles and expectations, index and firm level financial performance, and historical information. Instead of treating AI only as a technological development or only as a stock market phenomenon, the thesis examines how the economic expectations around AI are reflected in both market indices and company-level valuation measures using historical benchmarks. This provides a practical contribution for academics, investors, managers, and policymakers who need to interpret whether AI-related market enthusiasm is supported by observable fundamentals or whether it contains signs of speculative overvaluation.

1.3 Research Purpose

The purpose of this thesis is to evaluate quantitatively and using empirical market data and related information whether the current AI hype shows similarities to the historical market cycle of the dot-com bubble. This is done via the lenses of market-level and firm-level analysis. At the market level, the study compares the most representative indices across the two periods using visual similarities between the normalized index patterns, returns, volatility, statistical characterization, and similarity measures. At the firm level, the study compares financial multiples and profitability indicators across the eras of AI and dot-com. The aim is not to determine the intrinsic value of AI firms or to predict whether a future crash will occur. Instead, the thesis seeks to provide a historically grounded comparison of whether current AI-related valuation patterns resemble those observed during the dot-com period. Based on this purpose,

the main Research Question (RQ) of this thesis is:

> To what extent does the current AI hype resemble the dot-com bubble in terms of market-level cycle behavior and firm-level valuation patterns?

To answer this question, the thesis is guided by the following sub-questions:

1. To what extent do AI-era market indices resemble the dot-com period in terms of normalized price development, returns, volatility, and cycle similarity?
2. How do the valuation multiples and profitability indicators of AI-positioned firms compare with non-AI controls and with the dot-com bubble and control firms?

The hypotheses used to guide the empirical analysis are developed in Chapter 2 after the theoretical framework has been presented.

1.4 Scope and Delimitations

This thesis is limited to a historical comparison between the dot-com era and the current AI era. Even though insights regarding the evolution of the AI market cycle are obtained, the aim of this thesis is not to categorize the AI era as a bubble or not. Other historical bubbles, such as the Roaring Twenties or the housing bubble, are not considered. The market-level analysis is focused on major U.S. equity indices, especially technology-oriented indices that are relevant for comparing the dot-com and AI periods.

The firm-level analysis is limited to publicly available financial data for selected firms. The dot-com sample is affected by survivorship bias, because consistent financial-multiple data were difficult to obtain for firms that ceased to exist after the dot-com crash. As a result, the historical sample mainly consists of firms for which sufficient data could still be collected. The AI-era analysis is also limited by the fact that the current AI cycle is still ongoing, which means that recovery outcomes and full post-cycle performance cannot yet be observed.

The firm-level sample is also limited to selected publicly listed firms and does not cover the full population of dot-com or AI-era companies. Private AI startups and venture capital valuations are outside the scope of the study. In addition, the availability of financial measures differs across the two eras. The classification of AI-positioned firms is also based on available external AI-positioning data, and should therefore be interpreted as a practical proxy rather than a complete measure of AI exposure.

The thesis does not attempt to estimate firm valuation, nor does it build a predictive model of future market movements. The analysis is descriptive and comparative, using historical benchmarks, market indices, and financial multiples. It does not attempt to establish causal relationships between AI adoption and market valuation.

1.5 Thesis structure

The thesis is structured as follows. Initially, Chapter 2 presents the research overview and theoretical framework. It provides a comprehensive literature review on technology-driven bubbles, investor sentiment, AI-positioned firms, valuation challenges, and the use of financial multiples and market-cycle similarity measures.

Chapter 3 describes the methodology. More specifically, it explains the comparative research design, the market-level and firm-level data collection, the construction of the comparison groups, the financial measures used in the analysis, and the limitations of the used methodology.

Chapter 4 presents the results and the undertaken analysis. It first examines market-level index patterns across the dot-com and AI eras. It then presents the firm-level comparison using information regarding P/S, P/E, EV/EBITDA, EBITDA margin, and recovery years.

Chapter 5 discusses the findings in relation to the research question, hypotheses, and theoretical framework. It provides an interpretation of the results for both market-level and firm-level analysis.

Chapter 6 presents the conclusion of the thesis and outlines possible directions of future research.

Finally, the appendix contains supplementary material, including additional figures, data details, and supporting outputs that are relevant to the empirical analysis but not central to the main discussion.

2. Prior Work and Theoretical Framework

The objective of this chapter is to present a comprehensive literature review related to this thesis. Also, it aims to provide the necessary theoretical background for comparing the dot-com bubble with the current AI hype. More specifically, it initially reviews prior work on technology-driven bubbles, valuation uncertainty, and the role of investor sentiment in periods of rapid innovation. The chapter then connects these ideas to the two empirical perspectives used in this thesis, i.e., the market-level analysis of index behavior across market cycles and the firm-level analysis using the financial multiples of the firms os interest. In doing so, it provides the conceptual basis for why both broad market patterns and firm-level valuation measures are relevant to assessing whether the AI era resembles the dot-com period.

2.1 Technology-Driven Bubbles and Market Cycles

In this section, the literature review on the formation of how technology-driven bubbles and the motivation behind why market prices may deviate from firm fundamentals during periods of rapid innovation is presented. More specifically, the concepts of the market cycle (Britannica, 2026) and bubble (Kindleberger and Aliber, 2005) are introduced. Also, the concepts of technological diffusion (Wang, 2007) and diffusion evaluation gaps (Brynjolfsson et al., 2017; David, 1990) are discussed.

2.1.1 Overvaluation and the Anatomy of a Bubble

In general, a bubble is a vague term that lacks "a solid operational definition" (Wang, 2007). While a bubble is frequently described as a disconnection of the price of assets from their fundamental values, i.e., prices that "move too much to be explained by changes in the underlying fundamentals, such as dividends or cash flows" (Lansing, 2008), standard present-value models often fail to explain dramatic price shifts (LeRoy and Porter, 1981; Shiller, 1981). Baker and Wurgler (2007) argue that investor sentiment, defined as beliefs about future cash flows not justified by current facts, creates stock price patterns that defy explanation under the assumption of rational markets. This sentiment often centers on a "displacement" phase, where "perceptions of real improvements in the productivity" lead investors to "exaggerate the extent of the improvement in economic fundamentals" (Lansing, 2008). This exaggeration is viewed as a risk fueled by the "synergy between new and existing technologies" (Chang et al., 2016; Wang, 2007).

2.1.2 Technological Diffusion and the Diffusion-Evaluation Gap

A recurring theme in technological innovation is that "a wave of new firms implement the innovation and enter the market," but then market turbulence occurs because "if the innovation complements existing technology, some new entrants later will be forced out as more and more incumbent firms succeed in adopting the innovation" (Wang, 2007). This "shakeout" process explains why market valuations can be volatile even when the underlying technology is transformative.

Technological diffusion is the process by which an innovation spreads through a social and economic system (Rogers, 2003). Disruptive innovations (such as the Internet in the 1990s or AI today) act as external shocks that drive industry dynamics through the synergy between new and existing technologies (Wang, 2007). The so called "Diffusion-Evaluation Gap" is a significant offset which exists between the initial adoption of a technology and its realized economic benefits (Brynjolfsson et al., 2017; David, 1990). During early diffusion, markets often treat technology as a "platform for market shaping" rather than a mere product (Kaartemo and Nyström, 2021), leading to a "diffusion-evaluation gap." Because new high-tech firms often lack historical data, traditional valuation is difficult (Festel et al., 2013), and investors frequently "exaggerate the extent of the improvement in economic fundamentals" long before productivity gains materialize (Lansing, 2008).

2.1.3 Market Cycles and Bubble Phases

A market cycle is defined as the recurring phases of expansion, peak, contraction, and recovery in the financial market (Britannica, 2026). The driving forces behind the market cycles are the macroeconomic

fundamentals, credit conditions, innovation shocks, and investor sentiment. During the expansion phases, the pricing of assets is increasing due to the growth of earnings, liquidity, and the increased risk taking. When the peak phase is reached, there is an overvaluation of assets created by the speculative behavior of investors. Furthermore, the risk-taking is starting to fade due to compressed risk premia. The next phase is the contraction phase, where valuation corrections are taking place. The contraction phase takes place due to earnings disappointments, tightening financial conditions, and deleveraging. Finally, the following phase is the recovery phase, where phenomena such as capital reallocation, balance sheet improvements, observation of undervaluation opportunities, and the foundation of the next growth cycle are taking place.

For the case of bubbles, the previous phases of market cycles are further refined to the phases of early adoption (displacement or innovation shock), acceleration (boom), mania (euphoria), crash (panic), and consolidation (revulsion) (Kindleberger and Aliber, 2005). This refinement is in agreement with the Kindleberger–Minsky theoretical framework (Kindleberger, 1978; Minsky, 1986).

The market cycles driven by technological innovation tend to be more amplified compared to other types of market cycles. This is because of the higher growth opportunities, but also because of the higher uncertainty of the pricing of intangible assets. Note that this market-cycle framework supports **H1**, as defined in Section 2.6, because it suggests that if the AI boom follows a technology-driven bubble pattern, this should be visible in the behavior of market indices.

2.2 AI-Positioned Firms and Valuation Uncertainty

2.2.1 AI as a General Purpose Technology

In recent years, the advancements in fields such as computational power, data collection, cloud computing, signal processing, and AI algorithms have accelerated the commercialization of AI across a wide range of industries (Kumar, 2024; Shaik, 2024). Thus, currently, AI is posing as GPTech that has the potential to shape future economic and social relations (Stackpole, 2024). A GPTech, such as AI, is capable of creating productivity spillovers, reshaping competition, and altering the investment strategy of firms (OECD, 2025b). In financial markets, there is an increasing differentiation between AI-positioned firms and non AI-positioned firms, with the AI-positioned firms being valued higher due to anticipated growth (Brynjolfsson et al., 2021; Watanabe et al., 2022). In this thesis, the term AI-positioned firms corresponds to firms that adopt or implement AI as part of their processes, irrespective of which industry they belong to. Note that the level of AI-positioning differs among firms, as AI adoption varies across firms. Therefore, it is important to understand whether the impact of AI is justified with respect to future growth, or it is simply a speculative reaction following other previous speculative technological cycles (Pastor and Veronesi, 2009), or it is somewhere in the middle. The previous discussion provides the basis for the establishment and expectation for the valuation difference between the AI-positioned and the non-AI positioned firms. Note that this is later included in **H2** in Section 2.6.

2.2.2 Measuring AI-Positioning

In the AI era, the assessment of AI-positioning is a first step before evaluating whether current and expected future growth is reflected in valuation. In the literature, the assessment of the AI-positioning of a firm can be grouped into four main approaches. The first approach is based on text analysis of public disclosures, such as filings, earnings calls, job postings, and public announcements. For example, (Babina et al., 2024) relates the frequency and the use of AI-related terms in the public disclosures of firms to their strategic commitment to AI. The same article argues that the intensity of job postings requiring AI skills indicates stronger AI-positioning. The second approach uses patents and other innovation measures connected with AI. For example, the number of AI-related patents or patent filings, together with citation-weighted innovation measures, can reveal the level of AI adoption in a firm (Hall et al., 2005). The third approach connects AI exposure to investment in intangible assets, such as data, software, and organizational capabilities (Corrado et al., 2021). This is relevant because AI as a GPTech requires learning processes, new routines, skills, and organizational change before gains are realized (Brynjolfsson et al., 2021). Finally, the fourth approach assesses AI-positioning through market recognition, such as future growth options (Pastor and Veronesi, 2009), abnormal returns, or changes in valuation multiples (Alzeghoul and Alsharari, 2025; Henriksson and Nyqvist, 2022). As shown in Table 2.1, AI-positioning

can be assessed through technological capability, strategic commitment, human capital investment, and market recognition.

Table 2.1: AI-Positioning Assessment Dimensions

Category	Indicators	Reference
Technological Capability	AI patents; forward citations; R&D investment	(Hall et al., 2005)
Strategic Commitment	AI disclosure intensity in filings/calls; topic-modeling scores of AI themes	(Babina et al., 2024)
Human Capital Investment	AI-related job postings; AI skill intensity in workforce	(Babina et al., 2024)
Market Recognition	Valuation premia; abnormal stock returns following AI announcements	(Alzeghoul and Alsharari, 2025; Henriksson and Nyqvist, 2022; Pastor and Veronesi, 2009)

2.2.3 Valuing AI-Positioned Firms

Valuing AI-positioned firms is difficult because much of the expected value is connected to future growth rather than current accounting performance. In valuation models, AI may be reflected through higher expected revenue from new AI-enabled products (Sivathanu and Pillai, 2026), cross-selling opportunities (Sharma et al., 2022), margin expansion and scalability (Hyiamang and Liu, 2025), or higher long-term growth expectations (OECD, 2026). These effects are uncertain, which makes it difficult to determine whether high valuations are justified by future fundamentals.

Because of this uncertainty, market multiples are often used to compare AI-positioned firms with other firms and to assess whether an AI premium appears in market pricing (Moro-Visconti, 2024). Other valuation perspectives also help explain why such increases may occur. For instance, the Real Option Valuation (ROV) approach treats uncertain future projects as options that may create value even before they generate earnings (Myers, 1977). The framing of AI as a GPTech also suggests that firms may benefit from future innovation opportunities and spillovers (Bresnahan and Trajtenberg, 1995). In addition, accounting measures may underestimate intangible assets in knowledge-based firms (Lev and Gu, 2016), while technological innovation can increase both expected growth and uncertainty (Pastor and Veronesi, 2009). Therefore, an AI valuation increase may reflect both real growth opportunities and speculative market expectations. Table 2.2 summarizes these valuation perspectives that explain why AI-positioned firms may receive a market premium.

Table 2.2: Valuation approaches for AI-positioned firms

Origin of Value Creation	Academic Foundation	Practical Implication
Growth Expectations	Pástor & Veronesi (2009)	Higher Discounted Cash Flow (DCF) revenue growth
Intangible Capital	Lev & Gu (2016)	Multiple expansion
Innovation Output	Hall et al. (2005)	AI patent premium
Real Options	Myers (1977); Dixit & Pindyck (1994)	Platform optionality valuation
Sentiment Dynamics	Shiller (2000)	Thematic AI surges

2.3 Valuation Multiples and Speculative Markers

2.3.1 Valuation Challenges and Financial Multiples

The main difficulty in assessing and valuing recently appearing firms lies in the "lack of historical data and uncertainty" about their development that usually characterizes high-tech firms (Festel et al., 2013). Because traditional methods like Discounted Cash Flow (DCF) rely on highly sensitive "individual beta coefficients" (β coefficients) and subjective risk adjustments, market participants often default to multiples (Festel et al., 2013).

Fernandez (2023) notes that while multiples are "conceptually incorrect," they remain popular because they capture the market sentiment. This was evident during the dot-com era, where "stocks traded at multiples of earnings well in excess of historic levels" causing a decline in the "relationship between earnings and stock returns" which led many to question "whether traditional accounting and financial

information had lost its value relevance" (Morris and Alam, 2012). Griffin et al. (2011) attribute this "tech mania" to the behavior of institutional investors who rose technology stocks "more than five-fold" in the late 1990s.

2.3.2 P/S, P/E, EV/EBITDA, and EBITDA Margin

In contexts where earnings are volatile, very small, or negative, such as in the early-stage and high-growth technology firms, researchers often anchor themselves to these valuation multiples because they capture market sentiment during periods of high growth (Fernandez, 2023). This is particularly relevant in technology-driven cycles, where expectations about future growth can be reflected in market prices before the corresponding revenue or earnings have fully materialized (Fernandez, 2023; Lansing, 2008). Lansing (2008) notes that these sustained run-ups can represent "overreaction to technological innovation" where investors "exaggerate the extent of the improvement in economic fundamentals". The P/S ratio is useful in this setting because it relates market valuation to revenue and can still be observed when profitability has not yet matured (Fernandez, 2023; Morris and Alam, 2012). This makes it relevant for technology-oriented firms, where investors may price future scale and market share before stable earnings are achieved (Festel et al., 2013; Morris and Alam, 2012). The P/E ratio provides a complementary earnings-based perspective by comparing the market price with accounting earnings (Fernandez, 2023). However, in high-growth technology settings, P/E can be sensitive to small changes in earnings, especially when earnings are low or temporarily depressed (Fernandez, 2023; Morris and Alam, 2012).

In addition, the EV/EBITDA adds an operating-value perspective because EBITDA approximates the operational value, while enterprise value reflects the value of the whole firm. Morgan Stanley's discussion of valuation multiples emphasizes that enterprise-value multiples are useful because they relate total firm value to operating measures and allow comparison across firms with different capital structures (Mauboussin and Callahan, 2024b). The FABA study further treats EV/EBITDA as a ratio that can be compared between actual market values and a fundamental level derived from financial-performance indicators (Nenkov, 2025). This makes it relevant for assessing whether market pricing is supported by operating fundamentals (Nenkov, 2025). EBITDA margin complements this by showing whether operating profitability supports the valuation implied by the market. Together, P/S, P/E, EV/EBITDA, and EBITDA margin provide a firm-level framework for comparing market pricing, earnings, operating performance, and profitability across the dot-com and AI samples. This firm-level analysis method is used later in Section 2.6 to develop the hypotheses that compare AI-positioned firms with control firms and with the dot-com benchmark firms, especially with respect to hypothesis **H2**, **H3**, and **H4**.

2.3.3 Investor Sentiment and Valuation Plausibility

A secondary framework for identifying speculative cycles is to measure investor sentiment (Baker and Wurgler, 2007). More specifically, the authors of the previous paper argue that sentiment, i.e., the beliefs about future cash flows not justified by current facts, affects stocks that are "difficult to arbitrage," such as young, unprofitable high-growth firms (Baker and Wurgler, 2007).

When sentiment is high, traditional valuation models like DCF are often abandoned because they are too sensitive to "individual beta coefficient" and "uncertain factors about the future" (Festel et al., 2013). Instead, the market enters a phase of "uncontrolled risk" fueled by the synergy between new technologies and speculative capital (Chang et al., 2016).

To bridge the gap between historical cycles and the current AI boom, Albori et al. (2025) suggests an "agnostic valuation procedure" that assesses whether the expected growth of earnings required to justify current prices is "implausible given historical experience". The procedure assesses potential market overvaluation by deriving the "implied abnormal earnings growth rates" necessary to justify current stock prices. Unlike traditional models that rely on minimal subjective inputs or subjective forecasts, this approach uses a dividend discount model where parameters are constrained by equilibrium conditions and estimated using historical data through the generalized method of moments, and by computing the specific growth rates required to make current valuations compatible with rational pricing norms, the procedure evaluates the "plausibility" of the market's expectations (Albori et al., 2025).

By comparing the current technology sector - where the price of the US technology sector has more than doubled since 2023 - against the benchmarks of the 1990s, this framework identifies whether the

current "AI Hype" reflects real productivity gains or a departure from fundamental value (Albori et al., 2025; Lansing, 2008).

2.4 Market Cycles Using Similarity Measures of Indices

In general, technology-driven bubbles tend to behave similarly in terms of price evolution, following phases of financial exuberance, over-investment, and correction, despite the different driving technologies (Perez, 2002). This means that finding common patterns of evolution in representative indices may provide unique insights regarding the status of an ongoing market cycle and the possibility of the formation of a bubble. A similar support is also provided by the Kindleberger–Minsky framework (Kindleberger, 1978; Minsky, 1986). According to Minsky (1986), the bubbles are created endogenously due to transmissions between internal stages of financing, from hedge to speculative financing, and then to Ponzi financing. Furthermore, according to Kindleberger (1978), they follow phases which correspond to displacement, acceleration, mania, crash, and consolidation. Concurrently, stylized facts of asset returns from empirical finance, such as volatility clustering, fat tails, and asymmetric draw-downs, etc, empirically indicate repeating patterns across time, assets, and markets (Cont, 2001). In a similar way, the Phillips–Yu tradition (explosive dynamics and rational bubble framework) states that bubbles are identified, using econometric criteria, by explosive phases of price change (Phillips et al., 2015). Thus, the previous ideas can be used for similarity checks across time for similarities of market behavior.

Another theoretical framework that supports the cross-bubble similarity analysis is the techno-economic paradigm and innovation wave framework (Perez, 2002). More specifically, this framework suggests that a GPTech, such as the railways in previous technological revolutions (Perez, 2002) and currently AI (Agrawal et al., 2019; Brynjolfsson et al., 2021), creates a phase of financial over-investment, asset over-pricing, and market crashes before the realization of the promises of the new GPTech. The behavioral finance and collective belief framework can also be used for justifying this form of bubble analysis. Assuming that the investors are not always rational, they tend to follow trends, copy behaviors, and react to market stories without considering the economic fundamentals. The result of this is market prices that do not reflect the financial multiples (Kahneman and Tversky, 1979; Shiller, 2015). As similar behavioral mechanisms drive technology bubbles, this similarity is expected to be reflected to similar pattern of market evolution through time (Hong and Stein, 1999).

The narrative economics (Shiller, 2019) and semantic diffusion (Jackson and Yariv, 2007) framework can also be used for the analysis of bubbles through the mechanism of similarity check. More specifically, this framework explains how the technology driven bubbles are influenced and amplified by the propagation of stories and beliefs through media and investor communication. The technology sector is particularly sensitive to this phenomenon as it mainly includes firms whose valuation heavily relies on uncertain growth. Therefore, as similar propagation of stories emerges in different technological eras which may result in similar patterns over different eras, this means that the analysis of bubbles based on similarity measures is justified (Shiller, 2019).

The market structure and concentration-driven bubble framework is built around the idea that capital, attention, and future expectation are concentrated in a few dominant firms that heavily influence a sector or an index (De Loecker et al., 2020; Gennaioli et al., 2012). This is particularly the case to the technology sector where the concentrated market structure is influenced by the network effect (Katz and Shapiro, 1985), increasing returns, and the winner takes it all competition. Due to this structure, similarities in pattern evolution may arise in different eras, making the similarity analysis relevant.

The relative decoupling and fundamental divergence framework (Campbell et al., 1997a; Shiller, 1981; West, 1987) states that a bubble might be in the process of being formed when the pricing of an asset, sector, or index starts to diverge from the values of the underlying fundamentals for a significant amount of time (Campbell and Shiller, 1988). This is particularly the case to the technology sector where anticipated growth is isolated from current values of multiples. Consequently, similar disconnections of the pricing from the values of fundamentals might create similar patterns of evolution of indices over different eras. Thus, making the similarity analysis of the evolution of indices over different eras a useful tool.

Another justification for using similarity metrics for bubble analysis relies in the efficient market (Fama, 1991) versus inefficient market frameworks (Shiller, 2015). More specifically, the efficient market hypothesis attributes the pricing in the technology sector to the propagation of information and growth expectations. In contrast, the inefficient market hypothesis, following behavioral economics, justifies the

overpricing during a bubble to overreaction and herding. Both hypotheses may form temporal patterns of market evolution that can be consistent and thus compared over different eras.

The investor sentiment theory (Baker and Wurgler, 2006) is a useful tool for the analysis of bubbles in the difficult to price sector of technology as it explains the disconnection of the pricing from the fundamental values as a result of over-optimism, over-pessimism, or general mood. Due to the challenging pricing environment, the price may be significantly influenced by the general mood. Thus, comparative analysis using similarity metrics may be able to capture periods where the market evolution is heavily influenced by the mood of the investors.

The prospect theory claims that investment decisions are not always based on rationality, but it is the result of the evaluation against a reference point, such as previous pricing, expected returns, etc., (Barberis and Huang, 2008; Kahneman and Tversky, 1979). This theory is connected with the bubble creation because during bubbles, investors are focused on the potential large gains instead of the rational analysis. In addition, they overreact to losses instead of trying to objectively measure the fundamental values. For this reason, the similarity analysis might be able to capture repeated periods where the market evolution is based on risk-taking or overreaction to losses.

Finally, it is emphasized that the previous frameworks explicitly support **H1**, as given in Section 2.6. This is because the hypothesis tests whether the AI-era market indices show measurable similarity to the dot-com period.

2.5 Research Gap

A limited number of recent studies have directly compared the current AI boom with the dot-com era. (Dobre et al., 2020) compares the 1996–2001 dot-com period with the 2014–2018 AI landscape using stock-market data, but their analysis ends before the major acceleration of generative AI and large language models. Potrykus (2024) follows a partially similar approach to this thesis by comparing AI-related market indices with the corresponding dot-com indices using Dynamic Time Warping (DTW) and bubble tests. The study finds that the AI sector as a whole does not fully match the dot-com pattern, although some firms such as Nvidia and Meta show strong similarity. However, it focuses mainly on price patterns and does not include firm-level valuation and profitability measures. Subrahmanyam and Beainy (2026) examine whether AI is a real economic shift rather than another dot-com style bubble. Their argument is also partially close to this thesis because they consider financial fundamentals such as earnings, free cash flow, infrastructure investment, and adoption. However, this study used different multiples than the ones used by this study, and it mainly focuses on selected large AI firms and does not combine market-level similarity with a broader firm-level comparison.

Defend et al. (2026) provide a more recent comparison by analyzing AI and ex-AI portfolios against Technology, Media, and Telecommunications (TMT) and ex-TMT portfolios from the dot-com period, focusing on concentration, momentum, valuation, and risk characteristics. Chelikavada and Bennett (2026) also compare the two eras, but their study focuses mainly on scientific publishing activity, citation networks, and financial-market indicators as signals of hype-driven bubbles. Floridi (2024) argues that the AI era is another bubble that closely follows the dot-com era. This is done using theoretical arguments. Cao (2025) also compares the dot-com and AI booms, but from a theoretical perspective. The paper shows that AI may differ from the dot-com period if productivity effects spread faster across the economy. However, unlike this thesis, it does not provide a direct empirical comparison of market indices or firm-level multiples. A structured comparison of these studies, including their data, analytical approach, gaps, and conclusions, is summarized in Table B.1 in the appendix.

Although these studies are closely related to the topic of this thesis, they leave an important gap. Existing AI and dot-com comparisons mainly focus either on broad stock-market or portfolio behavior, or other indicators of hype such as research activity. They do not fully combine market-level cycle similarity with firm-level valuation and profitability evidence. As a result, they provide limited insight into whether AI-era firms are only benefiting from speculative market narratives or whether their valuations are supported by stronger operating fundamentals. This thesis addresses this gap by combining two complementary perspectives. First, it compares the dot-com and AI periods at the market level using normalized index performance, returns, volatility, correlation, and dynamic time warping similarity measures. Second, it compares the two periods at the firm level using P/S, P/E, EV/EBITDA, and EBITDA margin across dot-com bubble firms, dot-com controls, AI-positioned firms, and non-AI controls. In doing

so, the thesis extends previous AI and dot-com comparison research by connecting broad market-cycle behavior with firm-level financial fundamentals.

2.6 Synthesis and Hypothesis Development

The review of previous work in this chapter highlights two important and up to some point contradictory factors. On one hand, AI is framed as a GPTech with real growth opportunities, productivity effects, and strategic value. On the other hand, the review of the literature on technology bubbles, investor sentiment, and valuation uncertainty shows that markets may overprice expected benefits before they are visible in financial fundamentals. In addition, it is established theoretically that technological revolutions and GPTechs such as the railways, dotcom, and AI influence the market evolution in a way that similar patterns are formed between different technological revolutions and GPTechs. Due to the similarity between these patterns, it is possible to identify similarities of market evolution between different eras as expressed by the most representative measures. In this way, valuable information can be obtained about the present using similarity measures from past technological revolutions and GPTechs.

The previous contradiction along the provided theoretical foundation is the main motivation behind a study in which the AI era is assessed against the dot-com benchmark at both the index level and the firm level. The index level analysis aims to reveal characteristics of the general market evolution. In contrast, the firm level aims to assess the evolution of the fundamentals of a typical firm impacted by the GPTech of interest with respect to its price.

The hypotheses presented below are closely developed from the previous arguments. The market-cycle and similarity-measure discussion provides the basis for testing whether the AI-era indices resemble the dot-com era. The AI-positioning and valuation literature provides the basis for testing whether AI-positioned firms receive higher valuation multiples than comparable non-AI firms. Finally, the discussion of valuation multiples and operating fundamentals provides the basis for comparing the AI-era firm-level patterns with the dot-com benchmark. More specifically, the hypotheses of the current thesis are the following:

- **H1:** The AI-era market indices show measurable similarity, in terms of market behavior, to the corresponding indices of the dot-com era.

  This hypothesis follows from the numerous theoretical frameworks given in Section 4.1.2 and suggests that technology-driven bubbles may show recurring market-cycle patterns across different periods.

- **H2:** AI-positioned firms exhibit higher valuation multiples than their non-AI counterparts.

  This hypothesis follows from the literature on AI-positioning, which suggests that firms perceived as more exposed to AI may receive a valuation AI premium.

- **H3:** The statistical relation between the current AI-positioned companies and market valuation mirrors the patterns observed during the dot-com peak.

  This hypothesis follows from the discussion of speculative valuation multiples and the possibility that technology narratives can weaken the link between market prices and fundamentals.

- **H4:** AI-era firms exhibit similar operating profitability as the dot-com era firms.

  This hypothesis follows from the use of EV/EBITDA and EBITDA margin as indicators of whether market valuations are supported by operating fundamentals.

The previous sections have established why a study that includes both the market level and the firm level is necessary. In addition, the hypotheses that guide the empirical work are given along with their theoretical foundation. The next chapter turns from concepts to procedure by describing: a) the comparative research design; b) the way that dot-com and AI-era samples and data groups are constructed and sourced; and c) the analysis of the market and firm variables so that the results can be read against the framework and hypotheses given above.

3. Methodology

A comparative quantitative research design is used in this study to evaluate whether the current AI boom resembles the earlier dot-com technology market cycle. The methodology is structured around two complementary empirical perspectives. The first is the market-level perspective, where the evolution of the most representative indices and market-cycle patterns was compared. The second is a firm-level perspective where financial multiples and indicators were compared across dot-com era firms and AI era firms. Together, these two perspectives provided both a macro-level view of market behavior and a micro-level view of firm valuation and operating performance.

3.1 Research Design

In this thesis, a descriptive, comparative, non-causal, and historical design was used. Rather than evaluating AI-positioned firms in isolation, current valuation patterns were compared with historical market-cycle patterns from the dot-com era. More specifically, the dot-com bubble period was used as the primary historical reference of a technology-driven speculative cycle. The market-level analysis addressed whether broader AI-related market behavior resembled earlier patterns. However, it cannot capture elements of behavior in firm level. While, the firm-level analysis addressed whether similar valuation patterns could also be observed in the pricing of individual firms or between groups of firms carefully selected using a predefined criterion such as their AI positioning. The use of both levels is important to address the research purpose.

At the market level, a descriptive and comparative approach was followed. In more detail, the focus was on estimating the future fundamental values or market evolution. Instead, the evolution of the most representative indices across the selected periods was compared using normalized price changes, return and volatility patterns, and similarity metrics.

At the firm level, the same comparative logic was followed, but the focus was shifted from market indices to individual companies. Rather than estimating firm value, recorded financial multiples were compared across the defined firm groups. This allowed a direct comparison between the current AI era and the dot-com benchmark, using financial multiples commonly used by investors and researchers.

3.2 Data Collection

The data collection process followed the study's two-level structure of market-level and firm-level data.

3.2.1 Market-Level Data Collection

The market-level dataset was composed of the evolution of the most representative indices. As both the AI and the dot-com era are technology driven periods, the used data were drawn from the technological indices of NASDAQ Composite and NASDAQ 100. In more detail, the NASDAQ Composite is an index with a broad technology and growth-oriented market. Also, the NASDAQ 100 is an index dominated by large-cap, technology-heavy, innovation-intensive firms. Therefore, both indices are suitable to capture the effect of the GPTech of AI and the Internet. In addition, in order to capture the influence of the driving technology on the evolution of the general market, the SP 500 index was also considered.

For the AI-era, the collected data included the evolution of the NASDAQ Composite, NASDAQ 100, and SP 500 for the period between 2015 and 2026 (at the time of writing) (Brynjolfsson et al., 2021). In addition, for the dot-com bubble, the collected data corresponded to the period between 1995 and 2005 (Shiller, 2015). Note that the collected data included the daily adjusted close values of the previous indices. The data source is (TradingView, nd). The adjusted daily closing values were used because they can be used to calculate the normalized index evolution, daily log returns, rolling volatility, and similarity measures.

Finally, it is emphasized that as the two eras of AI and dot-com occur in different chronological periods with different macroeconomic conditions, index composition, interest-rate regimes, and market structure,

the provided analysis should be viewed only as a historical comparison rather than a causal test. More specifically, the collected index data, once processed as described below, are used to identify quantitatively measured similarities between the AI and the dot-com era. However, the identified similarities are projected to relevant financial and market information in an attempt to provide notions of causality.

3.2.2 Firm-Level Data Collection

The dot-com sample was assembled from a manually collected dataset of public firms identified as either bubble firms or control firms. The primary identification of these firms was derived from the Wikipedia Archive list of the *Companies Affected by the Dot-com Bubble* [1], which provided a comprehensive list of Pure-Play internet firms, hardware providers, and telecommunications companies that defined the 1990s technology boom. However, due to the historical nature of the 1998–2002 period, financial data were available for only a limited number of companies from that period. In addition, financial databases often presented only the surviving firms, omitting data for firms that were liquidated or delisted following the 2000 crash, which could have severely biased the collected data.

The AI-era sample was assembled from two main sources. The first was the Fortune-500 AI Adoption Index compiled by Salt Technologies AI (2026). This index identified 50 companies within the Fortune 500 that had demonstrated the most significant adoption of AI technologies across their business units. The second source was the AI tracker Larridin (2026), which not only listed AI-positioned firms, but also provided a multi-dimensional "AI Score" for each firm based on four key aspects: a) adoption, which is defined as "the depth of integration of AI within internal processes of the firm"; b) proficiency, which is the "technical maturity of the firm's AI deployments"; c) impact, which is defined as the "measurable influence of AI on the firm's value proposition"; and d) overall score defined as the "weighted aggregate of the firm's AI-positioning" Larridin (2026).

In addition to these scores, the data set incorporated high-frequency indicators of AI investment that are otherwise difficult to capture in traditional financial statements. These included AI hiring-post intensity, the count of active AI initiatives, and the number of strategic AI partnerships. Both sources were used to compile a dataset in which firms were classified as AI-positioned firms or non-AI controls. The classification of AI-positioned firms was based on the ranking values taken from Larridin's AI Tracker (Larridin, 2026). Firms with an overall AI score of 4.0 or above were classified as AI-positioned firms. The firms used as controls were selected from the same tracker but with the constraint that ranking values are between 1.3 and 3.9. This rule was applied to create a transparent and repeatable split between firms with stronger observable AI positioning and firms with weaker AI positioning. Those firms with scores below 4.0 were retained as non-AI controls because they show weaker AI positioning. Alternative cut-offs could have been used; the selected thresholds were preferred because they create a clear and replicable split between stronger and weaker AI-positioned firms within one external classification system.

The final dot-com sample contained 47 firms, consisting of 17 dot-com bubble firms and 30 dot-com control firms. These counts were largely determined by data availability. The starting list from Wikipedia was much larger, close to 100 firms, but many firms could not be retained because historical valuation data were missing, incomplete, or available only for a short post-bubble period. In addition, as discussed later in section 3.5, survivorship bias is likely, because firms that were delisted after 2000 are under-represented in available databases.

The final AI-era sample contained 102 firms, consisting of 60 AI-positioned firms and 42 non-AI controls. The AI-era counts follow the same sample-building logic as the dot-com groups, but under a more data-rich environment and easier to access. Larridin (2026) tracks hundreds of firms with publicly observable AI-related information, a larger contemporary sample could be assembled. The study does not claim that these 102 firms represent all AI-exposed listed firms. Rather, they represent the subset of tracker firms that met the classification thresholds and had usable valuation data.

For each firm, the collected variables included the company name, ticker, country, sector, incorporation year, valuation multiples, and profitability indicators. The main valuation variables were P/S, P/E, and EV/EBITDA. The main profitability variable was EBITDA margin. In addition, the dot-com dataset contained years-to-recovery values, which were used to analyze how long firms took to regain their prior stock price levels after the dot-com correction. Most of the firm-level valuation data for this sample were collected from the Macrotrends (2026) dataset, which provides both raw and pre-calculated valuation

[1] List of companies affected by the dot-com bubble, *Wikipedia*: `https://en.wikipedia.org/wiki/List_of_companies_affected_by_the_dot-com_bubble`

multiples and other related financial indicators for many firms. For the cases where a ratio was not directly available, it was calculated by its composing variable, as they were provided by the source of data. For example, the market capitalization and EBITDA were used to calculate the required values of EV/EBITDA. As shown in equation 3.1, all variables required for the calculation of EV/EBITDA on the right-hand side were provided by the source datasets. This is the case for both eras.

$$\text{EV/EBITDA} = \frac{\text{Market Value of Equity} + \text{Total Debt} - \text{Cash}}{\text{EBITDA}} \tag{3.1}$$

The time resolution of the variables differed by metric. In the AI sample, P/S and P/E were available for multiple snapshots: the yearly maximum values for 2024, 2025, and 2026, together with the last available observation as of April 2026. However, EV/EBITDA and EBITDA margin were available as single-period firm-level observations for the 2025 yearly snapshot and not 2026. This was because EBITDA is calculated on a full-year basis, and a comparable full-year 2026 EBITDA value was not yet available at the time of the analysis. Quarterly EBITDA could have been used instead, but equivalent quarterly data were not sufficiently available for the dot-com era.

In the dot-com sample, the comparable valuation points were the peak values of both 1999 and 2000, together with the 1999 and 2000 peak observations for P/S and P/E, and for EV/EBITDA and EBITDA margin, only the 1999 period observations were used. In the dot-com sample, the comparable valuation points were the peak values for 1999 and 2000, along with the peak observations for P/S and P/E in 1999 and 2000. For EV/EBITDA and EBITDA margin, only the 1999 period observations were used. Although 2000 data were available, the bubble burst in March 2000, and the full-year EBITDA for that year was therefore heavily affected by the post-peak. As a result, the 2000 EBITDA-based measures did not reflect the bubble period alone, but rather a mixed pattern of final accelerating stages and market crash stage. For this reason, the 1999 observations were considered more appropriate for capturing the EBITDA-based profile of the dot-com bubble period.

The data cleaning procedure followed several steps. First, column names were standardized to a consistent machine-readable format. Second, textual percentage fields were converted to numeric form. Third, rows with blank values were removed from the dataset. Finally, categorical group labels were created for the four analytical groups used throughout the thesis: a) dot-com bubble firms, b) dot-com control firms, c) AI-positioned firms, and d) non-AI controls.

3.3 Variable Construction and Measures

3.3.1 Group Definitions

The firm-level analysis was constructed based on four comparison groups: a) Dot-com bubble firms are firms classified in the historical sources as strongly exposed to bubble-like valuations; b) Dot-com control firms are firms from the same era that were not classified as bubble firms, and used as a comparison group; c) AI-positioned firms are firms in the AI-era dataset with a Larridin AI ranking of 4.0 or above; and d) Non-AI controls are firms from the same contemporary dataset that serve as the comparison group for the AI-positioned sample and were selected from Larridin ranking values between 1.3 and 3.9.

These group definitions were used consistently throughout the firm-level analysis. In the dot-com sample, a binary indicator was also used for determining whether a firm belonged to the bubble or control group. In the AI sample, another binary indicator was used to determine whether a firm was classified as AI-positioned or a non-AI control after the ranking-based threshold rule was applied. While firms differ in industry, size, and maturity, the comparison is designed as a benchmark analysis rather than a strict one-to-one matching, focusing on broad valuation patterns across comparable groups.

3.3.2 Financial Multiples and Indicators

The study used different financial multiples and indicators in the firm-level analysis, reflecting the different purposes of each analytical block. Four main measures were used in the comparative analysis: a) *Price-to-Sales (P/S)*: This ratio relates market value to revenue and is particularly useful in technology companies as firms may still be in a growth phase and earnings-based measures may be unstable; b) *Price-to-Earnings (P/E)*: This ratio relates market value to earnings and provides an earnings-based valuation perspective. It is informative, but also sensitive to very small or volatile earnings values; c) *EV/EBITDA*:

The EBITDA is a computation of a firm's earnings before interest, taxes, depreciation, and amortization are deducted (Berk and DeMarzo, 2024). The EV/EBITDA ratio relates enterprise value to operating earnings before interest, taxes, depreciation, and amortization. It provides a valuation measure that is closer to firms' operating cash-generation capacity; and d) *EBITDA Margin*: This profitability measure shows operating earnings relative to revenue and provides an indication of the operating performance underlying market valuations.

A fifth measure was used, which is the years to recover. This variable is available only for the dot-com sample and captures how long it took firms to recover their highest stock price levels after the bubble correction.

3.3.3 Descriptive and Comparative Indicators

The firm-level analysis relied primarily on descriptive statistics and comparative visualizations. For each metric, the analysis reported count, mean, median, standard deviation, minimum, and maximum values by group. The mean captures the arithmetic average, while the median is used as a more robust measure of central tendency in the presence of skewed distributions and extreme outliers. These two measures were reported together because the gap between them is itself informative and helps to understand the distribution of valuations within each group.

3.4 Analysis Procedure

3.4.1 Market-Level Analysis Procedure

In this thesis, the objective of the market level analysis is to find similarities of market evolution between the AI and the dot-com era, as reflected by the evolution of the most representative indices of the two eras. Practically, this can be done by identifying shared patterns of price evolution between the two eras or using other metrics that quantify statistical similarity. In more detail, in this thesis, the identification of common patterns between the considered indices was done using: a) a visual interpretation of the price evolution; b) statistical measures of similarity between the time series that represent the price evolution; and c) the distribution of volatility and returns. However, for obtaining consistent results and fair assessments of similarity, firstly, the raw data of evolution of the indices of interest were processed appropriately, and secondly, the appropriate measures of similarity were used.

Focusing on the identification of similarities using visual observations, for fair comparison between the two eras but also between different indices, the time series that represented the daily evolution of the indices were normalized so that day zero had the value of 100. This approach is aligned with (U.S. Bureau of Labor Statistics, 2023). In this way, it was possible to identify similar trends of evolution despite the different values of absolute pricing. Note that as there is a 10 year gap between the end of the dot-com era and the beginning of the AI era, there is a significant difference in the pricing of the indices in terms of absolute value due to the different market capitalization and the different general frameworks.

To gain a better insight into the similarity of the evolution of the indices of interest between the dot-com and AI eras, the correlation and DTW distance (Sakoe and Chiba, 1978) metrics were also used. Here, the correlation between the time series $\mathbf{x}$ and $\mathbf{y}$ is defined as the sample correlation given as:

$$r_{\mathbf{xy}} = \frac{\sum_{i=1}^{n}(x_i - \mu_{\mathbf{x}})(y_i - \mu_{\mathbf{y}})}{\sqrt{\sum_{i=1}^{n}(x_i - \mu_{\mathbf{x}})^2}\sqrt{\sum_{i=1}^{n}(y_i - \mu_{\mathbf{x}})^2}} \tag{3.2}$$

where, x_i, and, y_i, $i = 0, \ldots, L-1$ are the samples of the sampled normalized logarithmic time series; $\mu_{\mathbf{x}}$ and $\mu_{\mathbf{y}}$ are the sample means of the two time series. Also, the DTW distance is a metric of similarity suitable for two series of different length (Keogh and Ratanamahatana, 2005). The DTW distance is defined as the minimum cumulative alignment cost over all possible warping paths between two time series, where a possible path is defined as the path that satisfies the boundary, monotonicity, and continuity constraints (Sakoe and Chiba, 1978). The computation of the DTW distance in this thesis is done using the fastdtw module (Salvador and Chan, 2007; Salvador et al., 2016).

For being able to compute consistently the correlation and DTW metrics, the time series of the evolution of the considered indices, before their normalization they were adjusted to the same length. This was done by sampling the AI time series in order to have the same length as the time series of the

dot-com era. After normalization, the logarithm of the normalized series was obtained in order to ensure equal changes in the vertical axis corresponded to equal relative changes. As the focus was on the main trends and not in the short term noise, the LOWESS local regression method (Cleveland, 1979) was used to smooth the sampled normalized logarithmic time series of the evolution of the indices. In the next step, the time series of the considered indices were standardized via the operation of:

$$z_i = \frac{x_i - \mu_{\mathbf{x}}}{\sigma_{\mathbf{x}}}, \tag{3.3}$$

where, $\sigma_{\mathbf{x}}$ is the sample standard deviation of the time series; and z_i is the new standardized time series.

As discussed in Section 2.1.3, a bubble period (era) is composed by five phases, i.e., early adoption, acceleration, mania, crash, and consolidation (Kindleberger and Aliber, 2005). For the case of the dot-com era, the identification of the previous stages is readily available from the observation of its most representative indices. Thus, interesting observations can be drawn by exploring whether, for any time period, the evolution of the representative indices of the AI era presents similar patterns of evolution as the evolution of the corresponding indices of the dot-com eras for its different phases. For this reason, in this thesis, we deployed the concept of the daily rolling DTW distance and identified regions of evolution of the indices of the AI era that have a small DTW distance (high similarity) from segments of evolution of the most representative indices of the dot-com era that correspond to its different phases. In more detail, for the selected indices of the AI and dot-com eras and for each day of evolution of the AI, the DTW distance was computed between the segment of the dot-com era that corresponds to a specific phase with the segment of the AI evolution which includes the current day and a number of previous days that make the segment for the AI era equal to the segment of the dot-com era. Note that for obtaining consistent and interpretable results, the daily rolling DTW distance was applied to the logarithm of the normalized close values of the indexes of interest for both eras.

Furthermore, for identifying further similarities and differences between the two eras, an analysis that takes into account the relation between the volatility and returns of the most representative indices was provided. The objective of this analysis was to identify patterns of the price change, in terms of magnitude and direction, while, taking into account the dispersion of these returns. In order to achieve this, we expressed the price evolution of the considered indices in logarithmic scale. In the next step, we computed the (log) return of the i-th day using the logarithm of the prices evolution of the AI and dot-com era as:

$$r_i = \ln\left(\frac{P_i}{P_{i-1}}\right), \tag{3.4}$$

where, P_i and P_{i-1}, are the close value for the i-th and $(i-1)-$th day, respectively. Following this, the rolling volatility (risk) for the i-th day, which is defined as the sample standard deviation over a specific time window, was computed as:

$$\sigma_i = \sqrt{\frac{1}{L-1}\sum_{l=0}^{L-1}(r_{i-l} - \bar{r}_i)}. \tag{3.5}$$

Here, L is the length of the window in days, and $\bar{r}_i$ is the sample mean of the window of the returns computed as, $\bar{r}_i = L^{-1}\sum_{l=0}^{L-1} r_{i-l}$. Note that the window length in this study was 21 days. In the final step, the rolling volatility was annualized by being scaled by a factor of $K = \sqrt{252}$, where, 252 represents the days of trade during a year, as $\sigma_{i,\text{annualized}} = K\sigma_i$. The previous scaling was done for being able to: a) compare volatility over different periods, b) make the comparison between indices consistent, and c) provide a meaningful interpretation of magnitudes.

The outcome of the previous process is pairs of $(r_i, \sigma_{i,\text{annualized}})$, $i = 0, \cdots, L-1$, where, L is the length of the considered time series (evolution of the considered index). These pairs can be depicted by simple scatter plotting. Alternatively, more advanced plotting is possible via the use of the bi-variate KDE (Silverman, 1986). In more detail, the bi-variate KDE or simple KDE in this thesis, is a smooth non-parametric estimator of the joint density distribution of a bi-variate distribution. For the case of returns of pairs of $(r_i, \sigma_{i,\text{annualized}})$, $i = 0, \cdots, L-1$, the KDE of volatility and returns is defined as:

$$\hat{f}(x, y; H) = \frac{1}{L}\sum_{i=0}^{L-1} K_{\mathbf{H}}\left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} - \begin{bmatrix} r_i \\ \sigma_{i,\text{annualized}} \end{bmatrix}\right), \tag{3.6}$$

where, $K_{\mathbf{H}}(\cdot)$ is a bi-variate kernel function, while, $\mathbf{H}$ is a symmetric positive-definite bandwidth matrix controlling the amount and orientation of smoothing.

3.4.2 Firm-Level Analysis Procedure

First, a descriptive profile of the sample was compiled before any valuation comparisons, for each group. This profile recorded the number of firms included, summarized incorporation years through the median and range of founding years as a proxy for firm maturity, and described sector composition by identifying the leading industry categories within each group. The purpose was to provide context for the later metric comparisons and to assess whether the bubble, control, and AI-era groups differed systematically in sample size, age, or industry mix; the results are reported in Chapter 4 (Tables 4.2 and 4.3).

Second, each financial metric was analyzed separately. Grouped descriptive summaries were calculated for P/S, P/E, EV/EBITDA, EBITDA margin, and recovery years. For the two metrics with multiple AI snapshots, namely P/S and P/E, the analysis compared the AI-positioned and non-AI control groups across the 2024, 2025, 2026, and April 2026 observations. The dot-com sample was analyzed using the peak, 1999, and 2000 snapshots for the same two metrics.

Third, the presence of outliers was examined using the standard 1.5×Inter-Quartile Range (IQR) rule, also known as Tukey's fences (Irizarry, 2019; Tukey, 1977), where the IQR is defined as the difference between the third quartile and the first quartile, i.e., $\Delta Q = Q_3 - Q_1$. Here, Q_1 is the first quartile defined as the value below which 75% of the data falls, while, Q_3 is the third quartile, which is defined as the value below which 25% of the data falls. Observations lying below, $Q_1 - 1.5\Delta Q$, or above, $Q_3 + 1.5\Delta Q$, were identified as potential outliers. This step was used to document the presence of highly extreme observations and support the interpretation of differences between mean and median values in terms of sensitivity to outliers. Furthermore, comparative visualizations were produced for each metric. Logarithmic scaling was applied to reduce the visual dominance of outliers.

Finally, a recovery analysis was conducted for the dot-com sample only. In this thesis, a recovery analysis is defined as the analysis that compares the recovery duration of dot-com bubble firms and dot-com controls in order to examine whether firms associated with stronger bubble conditions also required more time to recover after the correction.

3.5 Reliability, Validity, and Limitations of the Collected Data

The methodology of this thesis has several strengths, but it also contains limitations that should be acknowledged. First, the study uses a comparative design in which the same core valuation concepts are applied across historical and current samples. This improves consistency and makes the results easier to interpret.

At the same time, the comparison between the AI and the dotcom eras is imperfect. The dot-com sample and the AI-era sample differ in firm age, sector composition, historical context, and sample size. For this reason, the analysis should not be interpreted as a strict, one to one, comparison between identical populations. Instead, it should be interpreted as a benchmark comparison between relevant groups.

With respect to the market level analysis, the reliability of the used data is ensured by the use of the adjusted closing values of well established indices. Also, the data were harvested from TradingView (nd) which is a well known and established source of data in the investment and finance industry. In order to establish consistency and replicability of the undertaken analysis, a well designed processing procedure was undertaken and carefully documented. More specifically, the data from the two eras were normalized for fair and direct comparison. Also, the same processing steps of logarithmic transformation, calculation of daily log returns, rolling volatility, LOWESS smoothing, and similarity measures were used among different eras and indices.

The validity of the undertaken market level analysis, the validity is limited as its objective is not identify causal connections. This means that the provided analysis identifies similarities between the two eras using different measures, but it does not identify the underlying economic mechanisms that created these similarities. In addition, the analysis solely focuses on public firms that compose the selected indices. This means that the obtained findings are not directly connected to private AI firms, venture-capital markets, non-U.S. markets, small-cap firms. For example, any influence that private firm OpenAI has on the AI era is not directly observable. Furthermore, the composition of the selected indices is changing over time. This means that the type of firms composing the selected indices during the AI and the dot-com era might be significantly different.

An important limitation is that the AI is still ongoing and its boarders between the different phases of a market cycle, according to the market (Britannica, 2026) and bubble Kindleberger and Aliber (2005)

frameworks, are not clear. In contrast, the dot-com era has undergone a complete bubble cycle. This means the provided comparison includes a level of uncertainty. Furthermore, the two eras experience different macroeconomic conditions, interest-rate regimes, monetary-policy environments, market structures, and levels of index concentration. Thus, the different indices are expected to have different behaviors in the two eras. Finally, the market level analysis provides market level findings, but it is unable to directly identify isolated firms that might be important.

With respect to the firm-level analysis, the data also contain strong skewness and several extreme outliers. To address this, the analysis reports medians alongside means and uses outlier-aware visualizations. Despite this, extreme observations remain an important part of the dataset.

Another limitation concerns the varying availability of data. More specifically, P/S and P/E are available across multiple AI snapshots, whereas, EV/EBITDA and EBITDA margin are only available as single-period values. Therefore, some comparisons are richer than others.

This study also depends on secondary datasets and historical reconstructions. This means that the reliability of the analysis partly depends on the quality of the sources from which the data were harvested. Also, the consistency of the historical reconstructions depends on the consistency of the historical records. While the recent AI era data is usually available and consistent across datasets, the dot-com era was usually missing or incomplete. The source used in this study was Macrotrends (2026),

An additional limitation concerns survivorship bias in the dot-com sample. It was difficult to obtain consistent firm-level data of financial multiples for companies that ceased to exist after the dot-com crash. As a result, the final historical sample consisted of firms that survived the 2000 bubble period and still exist today. This means that the dot-com benchmark may be a misrepresentation of the most fragile and extreme failed firms from that era, and the historical comparison should therefore be interpreted with this sample bias in mind.

In addition, in the firms grouping, the control groups comparability differs between the two eras. In the AI sample, comparability is relatively strong as both AI-positioned firms and non-AI controls come from the same tracker and use the same scoring dimensions. In the dot-com sample, comparability is more limited. Bubble and control firms are both public companies from the same historical period, but they were not drawn from an identical scoring system as the AI tracker. However, the cross-era comparisons could still be read as benchmark comparisons between eras, even though it is not a comparison between matched firm populations or similar classification trackers.

3.6 Ethical Considerations

While this research does not involve human subjects or clinical trials, it adheres to the ethical standards of financial research and data integrity.

Data Integrity and Transparency: A primary ethical concern in this study is the accurate representation of historical and current financial data. To reduce bias, the firm selection follows established indices, and all data cleaning processes—such as the removal of companies with missing data-are documented. This is to ensure transparency and to prevent the manipulation of the results to favor a specific hypothesis regarding market bubbles.

Intellectual Property and Licensing: The study respects the terms of service of all data providers. Financial data retrieved from public records or from paid subscription services is used for academic, non-commercial purposes only. All data sources are cited to acknowledge the intellectual property of the original compilers.

Objectivity in Analysis: Given the high level of public "hype" surrounding AI, the research maintained an objective stand to ensure that the final assessment of "overvaluation" is based on mathematical analysis results rather than subjective opinion or market sentiment.

3.7 Summary

This chapter provides a detailed description of how this study was undertaken. It describes the used comparative research design, the two parallel data sets (market indices and firm-level financial records). Also, it identifies the main limitations of the current thesis. Finally, it provided the main ethical consideration followed in this thesis. Finally, the following chapter presents the empirical results of this thesis.

4. Results and Analysis

This chapter presents and analyzes the empirical results of the study. In order to examine whether the current AI hype resembles earlier technology-driven dot-com market cycles, the analysis is conducted from two complementary perspectives: a historical, market-level perspective and a firm-level valuation perspective.

The first perspective, *Historical Market-Level Analysis: Indices Comparison Across Cycles*, aims to capture the broader market behavior. By comparing index patterns across periods, this part of the current thesis examines whether the current overall market behavior resembles the historically observed behavior during the earlier speculative technology cycles of the dot-com era.

The second perspective, *Firm-Level Analysis: Financial Multiples Comparison*, aims to examine whether the valuation patterns observed at the market level are also reflected in the pricing of individual firms. While index analysis shows how the market behaves in an aggregate way, it cannot by itself determine whether AI-positioned firms are priced differently from firms with low AI positioning, nor whether such differences are supported by financial multiples. The firm-level analysis, therefore, compares financial metrics across dot-com and AI firms. This makes possible the assessment of whether the observed AI hype is consistent with stronger fundamentals or instead reflects speculative overvaluation.

These two analyses are combined because they address different but related aspects of the research question. The index-level analysis provides a macro-level view of cycle behavior, whereas the firm-level analysis provides a micro-level view of valuation and operating performance. When considered together, they offer a comprehensive basis for evaluating whether the current AI hype resembles the earlier historical market bubbles of dot-com.

The current chapter first presents the historical market-level results from a cross-era comparison of the selected indices. It then turns to the firm-level results obtained using financial multiples.

4.1 Historical Market-Level Analysis: Indices Comparison

In this section, the objective is to explore whether there is a connection between the AI era and the well-known technology-driven dot-com bubble. This is done using different similarity measures between the AI and the dot-com era. The theoretical foundations for such a comparison are given in Section 2.4 and are based on the numerous theoretical frameworks presented there, ranging from the Kindleberger–Minsky framework (Kindleberger, 1978; Minsky, 1986), to behavioral finance and collective belief framework (Kahneman and Tversky, 1979; Shiller, 2015), and prospect theory (Barberis and Huang, 2008; Kahneman and Tversky, 1979). The interested reader is referred to Section 2.4 for a complete justification.

Initially, we use the joint normalized visual illustration of the daily evolution of the two eras to capture their visual differences and similarities. This form of analysis is grounded on the work of Perez (2002), who suggests that price evolution in different technology driven eras poses similar patterns over consecutive phases despite the different driving technologies. The transmission between different phases in periods of bubble is supported by numerous theoretical frameworks, such as the Kindleberger–Minsky framework (Kindleberger, 1978), behavioral finance and collective belief framework Kahneman and Tversky (1979); Shiller (2015), and the narrative economics (Shiller, 2019) and semantic diffusion (Jackson and Yariv, 2007) framework. The previous comparison is undertaken for the most representative indices of the eras we are studying.

In the next step, the statistical measures of correlation and DTW distances are utilized in order to show how close the trends of the considered indices are. Especially, the correlation comparison is motivated by the Phillips–Yu tradition (Phillips et al., 2015) which connects (identifies) bubbles with econometric criteria over explosive price changes. Similar motivation can be found in the Phillips–Yu tradition (Phillips et al., 2015) for the calculation of the DTW distances between two indices from different eras. This is based on the fact that previous processing is an econometric procedure. Here, a trend is a smoothed curve that captures the main direction of the index of interest using a window of specific length. An example of a trend is the SMA or LOWESS. Note that the trends in this section are generated using the LOWESS.

In order to obtain a suggestion if the AI era has already experienced any of the usual phases of a bubble,

i.e. early adoption, acceleration, mania, crash, or consolidation, as given in the Kindleberger–Minsky framework Kindleberger (1978); Minsky (1986), we make use of the notion of rolling DTW distance. More specifically, we calculate the rolling DTW distance of the daily evolution of the logarithm of the normalized close values of the index of interest for the period of 2017 to 2026, which corresponds to the AI era, with the segment of the daily evolution of the logarithm of the normalized closing values of the same index, corresponding to a given phase of the dot-com bubble. Several frameworks from Section 2.4, such as the techno-economic paradigm and innovation wave framework (Perez, 2002), the narrative economics (Shiller, 1981) and semantic diffusion (Shiller, 1981) framework, prospect theory (Kahneman and Tversky, 1979), etc, theoretically support such an analysis.

Finally, the relation between the rolling volatility and the logarithm of the returns of the index of interest is examined. More specifically, for both the AI and dot-com era, the scatter of the rolling volatility-log returns is shown. For better visualization, we include the KDE of rolling volatility-log returns in the same figures. Based on these visual illustrations, we can see if the volatility-log returns of the AI era have the characteristics of any of the considered historical technology-driven bubbles. This analysis is based on the stylized facts of asset returns from empirical finance (Cont, 2001).

In the following subsections, the previous analyses are undertaken for the market circles of the AI-era (2015 to 2026) (Brynjolfsson et al., 2021) and the dot-com bubble (1995 to 2005) (Ofek and Richardson, 2003; Shiller, 2015). For both cases, we consider as the most representative indices the: a) NASDAQ Composite (IXIC), b) NASDAQ 100 (NDX) and c) SP 500 (SPX) (Bae et al., 2024; Fama and French, 1992; Nasdaq, Inc., 2018; Ofek and Richardson, 2003). The main rationale for selecting the NASDAQ Composite and NASDAQ-100 indices is that they are technology and innovation-intensive. SP 500 is selected because it serves as a broad market benchmark. This allows comparisons between sector-specific bubbles and the overall market.

4.1.1 Normalized Evolution of the AI and Dot-com Eras

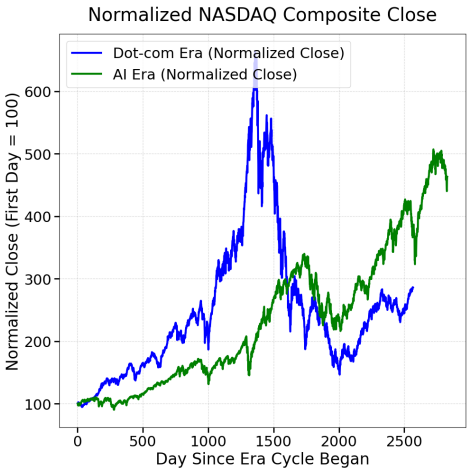
Initially, in Fig. 4.1, we present the daily evolution of NASDAQ Composite, NASDAQ 100, and SP 500 from the beginning of the cycle of the AI and dot-com eras (day 0) until the end of the dot-com era in day 2518 (2005) and until day 2793 (April 2026) for the the AI era. Note that, in all sub-figures of Fig. 4.1, the indices are normalized such that their value on day 0 is 100. This is done in order to capture and compare the movement trends of the considered indices without being constrained by their absolute values. This allows for a relative performance comparison in which trajectory, momentum, and draw-down behavior are highlighted as theoretically established in Section 4.1.2. Because there is a time difference of about 10 years between the end of one era and the beginning of another, their absolute values differ significantly. This can be seen in Appendix A, where the absolute evolution of the previous indices over time is shown.

The observation of Fig. 4.1 shows that during the dot-com era, the technology-heavy indices of NASDAQ Composite and NASDAQ 100 show explosive movements between days 1000 and 2000. This implies that the two technology indices during this period were in a highly speculative regime as described by the Phillips–Yu tradition (Phillips et al., 2015). In contrast, the same indices during the AI era show a more gradual, over time spread, and upward regime. This implies less extreme behavior and smoother trend reversals than in the dot-com era, suggesting a stronger connection to fundamentals. With respect to volatility and draw-down severity, we can see that the dot-com era is more dramatic. This is explained by the stylized facts of asset returns from empirical finance (Cont, 2001). On the other side, the AI era experiences corrections and retracements which are similarly deep, followed by new advances. It is worth noting that the deepest corrections and retracements during the AI can be attributed to external events, *i.e.* COVID, wars in Ukraine and Iran, and the trade-war rhetoric of President Trump after his election.

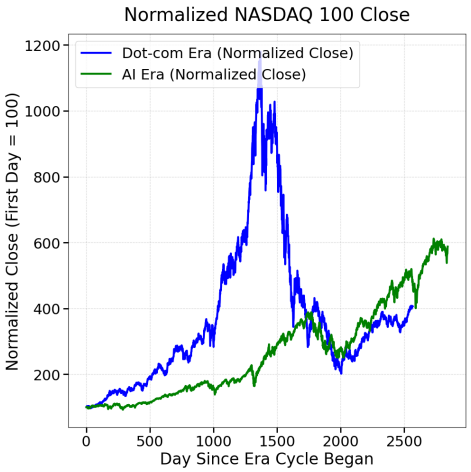
Focusing on the NASDAQ 100 index, which is the most concentrated because it includes large technology firms in both eras, we can see that concentration risk is higher in the dot-com era. This can be clearly seen when comparing the peaks of the NASDAQ Composite and the NASDAQ 100 indices. It can be seen that the NASDAQ 100 has almost double the relative advance compared to the NASDAQ Composite during the dot-com era, whereas during the AI era this advance is limited only to about 20%. This observation along with the market structure and concentration-driven bubble framework (De Loecker et al., 2020; Gennaioli et al., 2012), described in Section 4.1.2, indicates that the AI era has weaker concentration characteristics than the dot-com era, despite the presence of Magnificent Seven mega-caps which correspond to approximately 50% of NASDAQ 100 according to Kokkinidis (2025).

In Fig. 4.1, the SP 500 provides insight into the evolution of the wider market. This is because the SP 500 includes firms from different sectors. It can be seen that even though SP 500 follows the same pattern as the NASDAQ Composite and the NASDAQ 100 in both eras, there are major differences. Firstly, the peak relative performance of SP 500 is significantly lower than the other indices. And secondly, the peak of the relative performance of SP 500 of the AI is higher than the corresponding peak of the dot-com era, which is not the case for the other two indices. This suggests that the upward evolution during the AI era is not constrained to the AI technology sector but is more widely diffused across the market as predicted by (Rogers, 2003). This is also supported by the fact that the performance difference between the peaks of the considered NASDAQ indices and the SP 500 is wider during the dot-com era. This, combined with the longer upward time evolution, might indicate more sustainable development. This position is also enhanced by the fact that since the beginning of the AI era in 2015, the systemic environment (persistent low rates, especially during 2015-2021; large monetary expansion; passive investing and index funds; corporate buybacks; inflation and nominal growth; governmental support after COVID) supports the upward evolution.

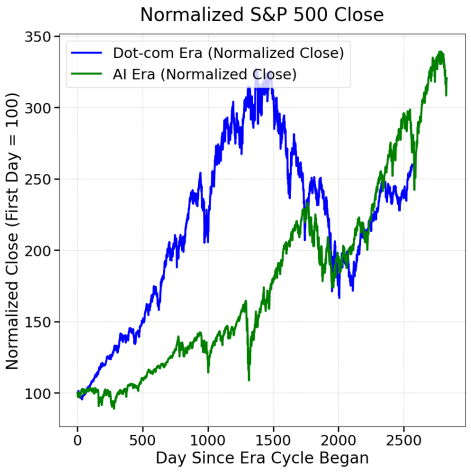
Another observation from Fig. 4.1 is that, for the case of the dot-com era, the corrections of the indices after major drops do not happen in the near future (inside the dot-com cycle). In contrast, for the case of the AI era, the corrections happen soon after a major drop within the era of AI. In addition, it is worth noting that all major drops during the AI occur due to external reasons. More specifically, the first major drop of the AI era happened in 2020 (around day 1400) due to Covid. The second major drop happened in 2022 (day 2000) due to the war in Ukraine. The third major drop happened in 2025 (day 2500) because of the trade-war rhetoric of President Trump.



(a) NASDAQ Composite



(b) NASDAQ 100



(c) S&P 500

Figure 4.1: Normalized Index Performance: Dot-com vs AI Era

4.1.2 Similarity Measures Between the AI and Dot-com Eras

To better understand the similarities and differences between the dot-com and AI eras, we present an analysis based on statistical measure of correlation and the time-series alignment metric of DTW distance. More specifically, in Table 4.1, we present the correlation and DTW distance for the evolution of the NASDAQ Composite, NASDAQ 100, and SP 500 indices. For both similarity metrics, the evolution of the indices of the AI era is sampled to ensure equal length. Also, the trends are generated using the LOWESS local regression method (Cleveland, 1979) on the standardized logarithmic values of the normalized evolutions of both eras. This is done to focus on the underlying shape of the trend rather than on secondary trends that can be considered as noise.

With respect to correlation, Table 4.1 shows values of correlation between the dot-com and AI eras of between 0.59 and 0.63. This indicates a moderate positive correlation. This means that the two eras share the same directional pattern across trends. However, since the correlation is significantly less than 1, the AI era cannot be interpreted as a repetition of the dot-com era. In contrast, this level of correlation can be explained by clear differences in terms of timing and shape. To isolate the timing dimension to some extent, Table 4.1 presents the DTW distance values for the two data sets. Note that the DWT distance is normalized in order to take values between $(0, 1]$, where smaller values of DTW correspond to a higher level of similarity. From Table 4.1, it can be concluded that all indices have a moderate level of similarity once the timing differences are not taken into account. Furthermore, SP 500 shows the highest similarity between the two eras.

Table 4.1: Similarity Measures Between Dot-com and AI Era Market LOWESS Standardized Trends

Index	Corr.	DTW
NASDAQ Composite	0.59	0.47
NASDAQ 100	0.61	0.49
S&P 500	0.63	0.40

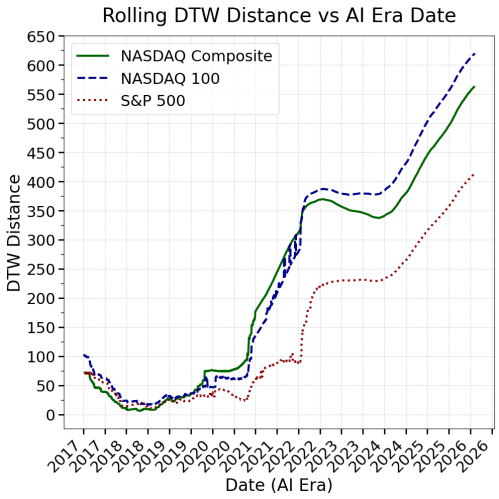
4.1.3 Index Similarity of the AI era with the Composing Phases of the Dot-Com Bubble

For understanding whether the indices of NASDAQ Composite, NASDAQ 100, and SP 500, during the AI era (2016–2026), exhibit evolution characteristics that are similar to the phases of early adoption (1995–1997), acceleration (1998), mania (1999–March 2000), crash (March 2000–2002) and consolidation (2003–2005) of the dot-com bubble, in Fig. 4.2, we present the rolling DTW distance of the whole evolution of the previous indices between 2016 and 2026 and the corresponding segments of the evolution each phase of the dot-com era. Note that the values of DTW distance in Fig. 4.2 are not normalized and can take values between 0 and infinity, with smaller values of DTW distance indicating a higher level of similarity.

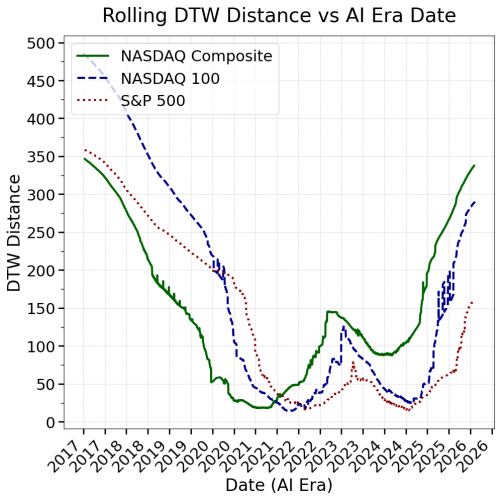
The observation of Figs. 4.2a to 4.2e shows that the similarity level (values of the DTW distance) over time is varying. This implies that the evolution of the indices during the AI era exhibits varying degrees of similarity across the phases of the dot-com era. In addition, the NASDAQ 100 has the highest DTW distance across all five phases in terms of peak value. Also, it usually has either the highest or second-highest DTW distance throughout time. Especially, this is the case for the similarity measure with the mania and crash phases of the dot-com era. This suggests that the highly concentrated, technology-heavy NASDAQ 100 index resembles the dot-com era phases the least, compared to the broader indices of the NASDAQ Composite and the SP 500. Also, across all phases, the SP 500 and the NASDAQ Composite indices usually have the lowest DTW distance (highest similarity). This indicates that, within the given time windows, the broader market (SP 500) or the broader technology sector (NASDAQ Composite) exhibited a pattern similar to the phases of the dot-com era, rather than the NASDAQ 100, which is dominated by mega-cap technology firms.

Let us now focus on each subfigure of Fig. 4.2 (comparison with the corresponding phase of the dot-com era). Initially, in Fig. 4.2a, we can see that until the end of 2020 for the NASDAQ 100 and NASDAQ Composite indices and the middle of 2020 for the SP 500 index, the DTW distance is consistently below 100. This implies that, during these time windows, the evolution of the indices resembles the early adoption phase of the dot-com era. However, as we move away from these windows, the DTW distance increases sharply, implying a sharp reduction of similarity. Therefore, it can be concluded that the considered indices exhibited similar patterns of evolution as the early adoption phase of the dot-com era only during the first years of the AI era, which were referred to earlier as the early adoption phase.

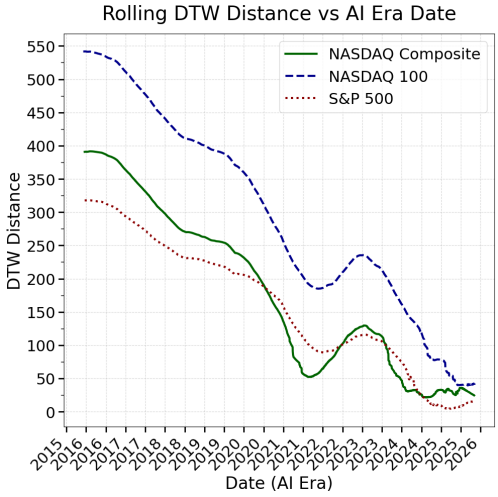
The comparison of the similarity of the AI era with the acceleration phase of the dot-com era is given in Fig. 4.2b. It can be seen that the high values of DTW distances (no similarity) in the early years of the AI era are reduced sharply until: a) NASDAQ Composite achieves a minimum in the second part of 2021; b) NASDAQ 100 achieves a minimum in the first part of 2022; and c) SP 500 achieves a minimum in the second part of 2022. In all cases, the minima have a value of DTW distance below 50, implying a high similarity to the acceleration phase of the dot-com era. After this point, the DTW distances increase again, then drop until they reach local minima at the end of 2024 (NASDAQ Composite) and at the beginning of 2025 (NASDAQ 100 and SP 500). Note that only NASDAQ 100 and SP 500 have a local minimum value of the DTW distance below 50. After these local minima, the DTW value increases sharply. This indicates that the considered indices resemble each other closely, in terms of evolution, during the acceleration phase of the dot-com era in two time periods. A plausible interpretation for these two periods, which include the minima, is that they constitute a single acceleration phase, interrupted



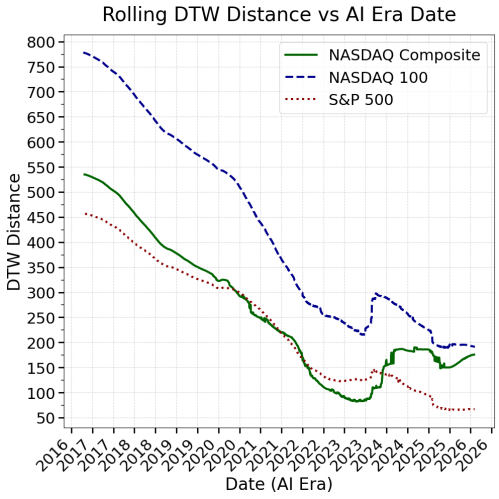
(a) Similarity with the early adoption phase of the dot-com era.



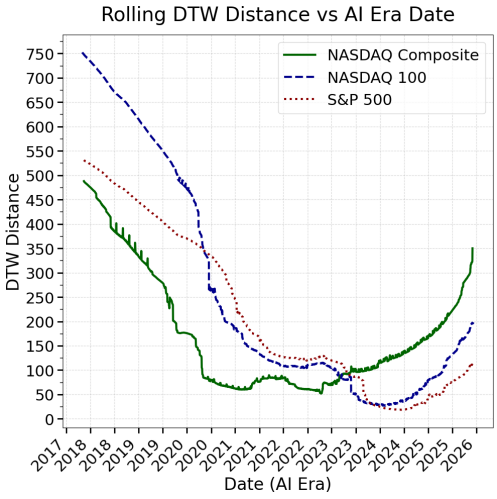
(b) Similarity with the acceleration phase of the dot-com era.



(c) Similarity with the mania phase of the dot-com era.



(d) Similarity with the crash phase of the dot-com era.



(e) Similarity with the consolidation phase of the dot-com era.

Figure 4.2: Rolling DTW distance between the AI era (2016 to 2026) and the segments of the dot-com era that correspond the the phase of early adoption (1995–1997), acceleration (1998), mania (1999–March 2000), crash (March 2000–2002) and consolidation (2003–2005) for the indices of NASDAQ Composite, NASDAQ 100, and SP 500.

for some time by the external factor of the beginning of the war in Ukraine (February 2022).

Focusing on the comparison of the evolution of the considered indices during the AI era with the mania phase of the dot-com era in Fig.4.2c, it can be seen that the indices follow a decreasing trend in the value of the DTW distance from the initial high value in 2016 up to the relatively small values mainly after 2025. An exception to this is a period between 2022 and 2023 during which the trend is reversed. However, after 2023, the trend reverses again downward. It should be noted that the DTW distance for all indices converges to values below 50 after 2025. These values of DTW distance indicate high resemblance with the mania era of the dot-com era. Also, the resemblance is not limited to a single section but is seen in both the technology-heavy and broad market indices.

The similarity between the evolution of the considered indices during the AI era and the crash phase of the dot-com era is shown in Fig. 4.2d. This figure shows that, initially in 2016, all indices have a high DTW distance (no similarity), which declines over time. More specifically, NASDAQ Composite and NASDAQ 100 reach a minimum value between 2023 and early 2024, while, at the same time, SP 500 reaches a local minimum. After that point, the DTW distance increases shortly until NASDAQ Composite and NASDAQ 100 attain a local minimum, whereas SP 500 does not seem to reach a local minimum in the considered time period. The first minima points are connected in time with the war in Ukraine, while the second minima points are connected with the trade-war rhetoric of President Trump. From the whole evolution of the DTW distance curves in Fig. 4.2d, it can be concluded that there is a crash in the AI era, analogous to the dot-com bubble, in terms of the shape of market evolution, in 2023 and 2025, but only to a partial extent.

Finally, in Fig. 4.2e, the DTW distance between the AI era and the dot-com-era segment corresponding to the consolidation phase is shown. The observation of this figure shows that all indices follow a υ shape. For the case of the NASDAQ Composite index, the minimum values of the υ shape extend from 2020 up to 2023. In contrast, for the cases of the NASDAQ 100 and SP 500 indices, the minimum values of the υ shape converge and extend between 2023 and 2025. This indicates that for the specific period, the evolution of the AI era followed closely the consolidation phase of the dot-com era. However, as we move closer to the recent period in 2026, this resemblance is steadily weakening.

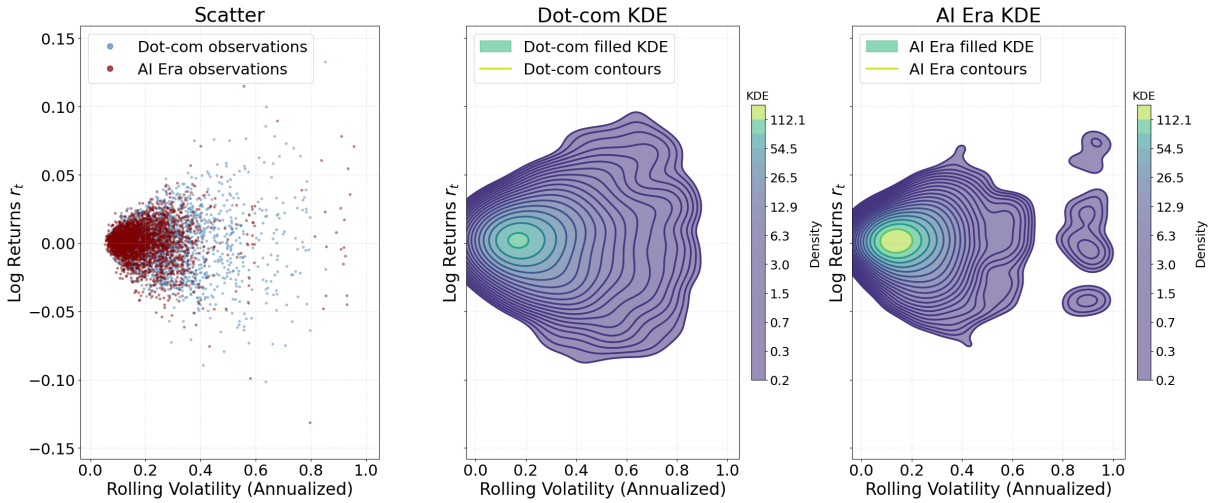
4.1.4 Volatility and Return Comparison between the AI and Dot-com Eras

In this subsection, the aim is to compare the AI and the dot-com era in terms of volatility and returns. This is done in order to capture the magnitude and direction of the price change while accounting for the dispersion of these returns.

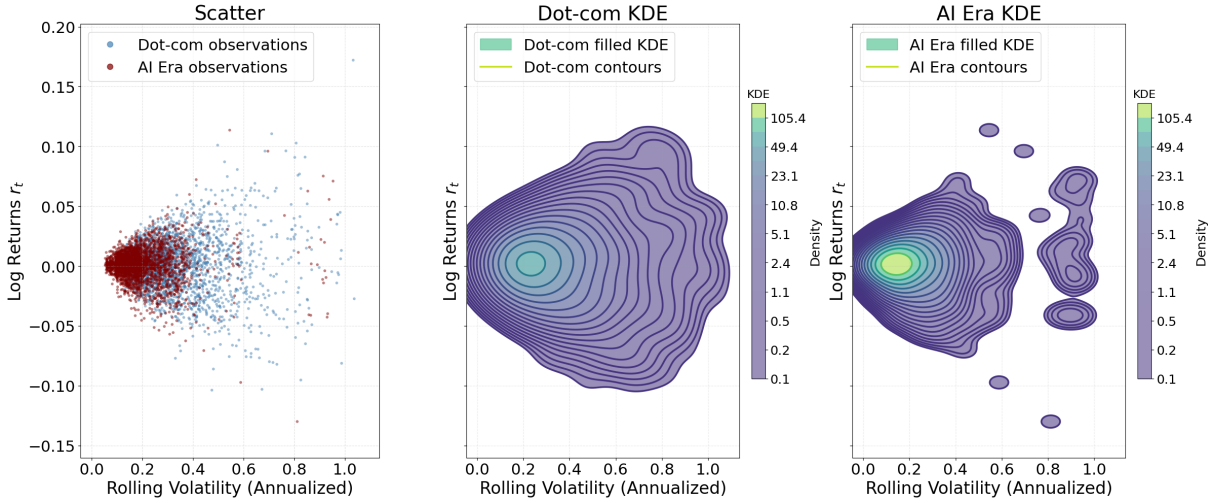
Figure 4.3 presents the connection between the rolling volatility (annualized) versus the (log) returns for the indices of interest (NASDAQ Composite, NASDAQ 100, and SP 500), both for the dot-com and AI eras. This is done using the scatter plot, KDE, and contour lines. More specifically, Figs. 4.3a to 4.3c correspond to NASDAQ Composite, NASDAQ 100, and SP 500, respectively, with the left figure presenting the the scatters of both eras, whereas, the center and right figure presenting the KDE and the contour lines of the dot-com and AI, respectively.

The joint observation of Figs. 4.3a to 4.3c shows that the region with the highest density of points of (log) returns-rolling volatility (annualized) is the low to moderate volatility (around 0.2) and around zero returns. Another conclusion drawn from the joint observation is that the two eras resemble each other in central distribution, but differ significantly in dispersion, tail shape, and clustering in the extreme-volatility regime. Focusing on the figures representing the KDE of the probability distribution of the data, we can see that the KDEs of the dot-com era has a wider and more continuous spread compared to the AI era. This is especially true for the NASDAQ Composite and NASDAQ-100 indices. In contrast, the AI era has a more concentrated spread of KDE combined with small regions of very high rolling volatility (annualized). From the previous observation, it can be inferred that the dot-com era has a wider and more continuous distribution of risk. Whereas the risk of the AI era seems to be concentrated mainly in the low-to-moderate regime, with small remote instances of very high risk (volatility).

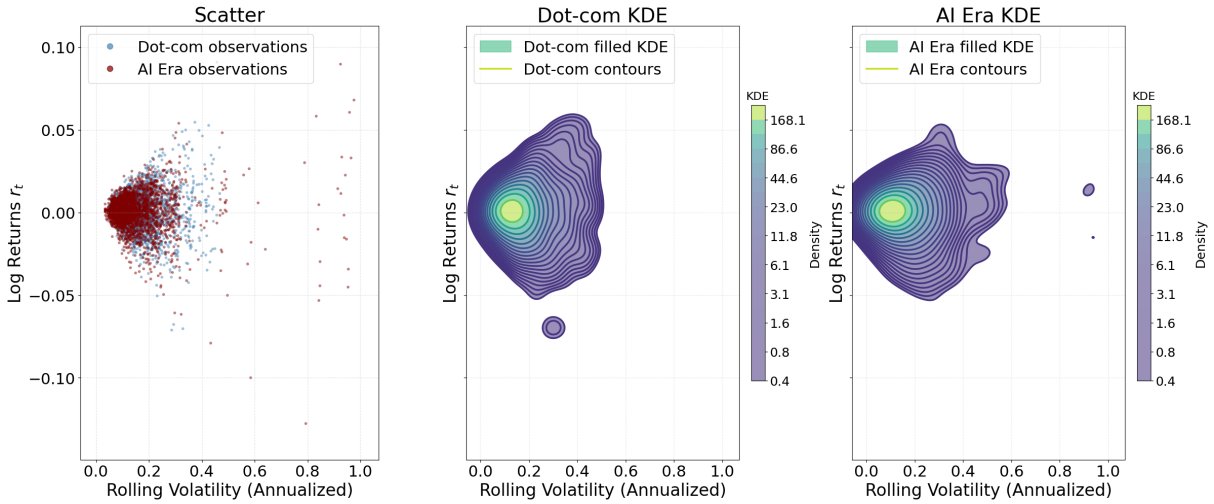
In Fig. 4.3a, it can be seen that the scatters are mainly positioned around zero returns and volatility between 0.1 and 0.25. In the dot-com era, we also see that the scatters extend into a higher-volatility regime with higher density than in the AI era. However, the AI era presents a number of outliers that experience a very high regime of volatility. With respect to the KDEs figures, it can be seen that during the dot-com era, the NASDAQ Composite exhibits a broad, continuous shape, spanning from low to high volatility and higher amplitudes of positive and negative returns. In contrast, the AI era has a denser



(a) NASDAQ Composite



(b) NASDAQ 100



(c) S&P 500

Figure 4.3: Returns and rolling volatility (annualized) for the indices NASDAQ Composite, NASDAQ 100, and SP 500 using scatters, the KDE of the dot-com era, and the KDE of the AI era.

KDE concentration in the same region as the dot-com era, but its main part extends to higher volatility levels (lower than in the dot-com era). In addition, there are three isolated regions of very high volatility. These regions suggest that there are short periods of bursty high volatility, which in some cases are combined with higher than usual returns.

The behavior of the NASDAQ 100 in terms of the relation of (log) returns and rolling volatility (annualized) is shown in Fig. 4.3b. It can be seen that this index has the widest spread among the considered indices. Also, mainly the KDEs figures, but also the scatters, follow the same patterns as in the case of the NASDAQ Composite, but with two differences. Firstly, the dot-com era has a weaker central regime compared to the AI era. In addition, it extends to higher volatility (positive and negative) than the NASDAQ Composite. And secondly, the AI era exhibits more isolated islands of high volatility at different levels. Furthermore, Fig 4.3b provides, in terms of distribution of scatters and KDE, the strongest evidence of the difference between the dot-com and AI era.

In Fig. 4.3c, it can be seen that the SP 500 index has the most compact and concentrated distribution of returns and volatility, both in terms of scatters and KDEs, among the considered indices. It can be seen that both eras have a high concentration in the central regimes (regions with volatility close to 0.2). However, the dot-com era has a wider distribution towards the higher volatility, while the AI era has outliers of very high volatility with different levels of (positive and negative) returns. Note that the previous phenomena have a weaker extent compared to the NASDAQ Composite and NASDAQ 100 indices. From the above analysis, it can be implied that the SP 500 has a more stable and compact distribution of volatility and returns compared to other NASDAQ Composite and NASDAQ 100 indices, in both eras. From this, it can be inferred that AI-related behavior has a smaller effect on the general market than on the technology-oriented indices, the NASDAQ Composite and the NASDAQ 100.

4.2 Firm-Level Analysis: Financial Multiples Comparison

This section presents the firm-level results of the study by comparing financial multiples across dot-com bubble firms, dot-com control firms, AI-positioned firms, and non-AI controls. The purpose is to examine whether the valuation patterns observed at the market level are also visible in the financial metrics of the individual firms.

4.2.1 Descriptive Profile of the Compared Firms

Before comparing valuation multiples across eras, it is useful to describe the basic profile of the firms included in the sample. This provides context for the comparisons and helps clarify the distribution of the groups across other factors, such as firm age and sector.

The final firm-level sample consists of 17 dot-com bubble firms, 30 dot-com control firms, 60 AI-positioned firms, and 42 non-AI control firms. Within the dot-com sample, bubble firms appear younger than the control group. The median incorporation year for dot-com bubble firms is 1994, compared to 1973 for dot-com control firms. A similar comparison within the AI-era sample shows that AI-positioned firms are also a bit younger than their control group, with a median incorporation year of 1976 compared to 1961 for non-AI controls. At the same time, both AI subgroups are more mature overall than the dot-com bubble firms, reflecting that stronger and more established firms are forming the sample of the AI era. The detailed composition of the considered groups of firms is given in Table 4.2.

Table 4.2: Sample size and founding-year profile of compared firm groups

Group	Count	Median year	Min year	Max year
Dot-com bubble	17	1994	1876	1998
Dot-com controls	30	1973	1839	1999
AI-positioned	60	1976	1784	2019
Non-AI controls	42	1961	1839	2011

As shown in Table 4.3, the sample composition also differs between the groups compared. The dot-com sample is relatively narrow and concentrated in software, internet retail, communications, and semiconductors. Within the AI-era sample, both AI-positioned firms and non-AI controls are strongly

represented in technology, but the surrounding sector mix differs slightly. These patterns suggest that the AI comparison is not simply between one homogeneous "AI sector" and one homogeneous non-AI group, but rather between broader sets of firms distributed across sectors. This observation is in agreement with the characterization of AI as a GPTech (Stackpole, 2024) and the concept of technological diffusion (Rogers, 2003).

Table 4.3: Largest sector categories within each compared firm group

Group	Sector	Company count	Share of group (%)
Dot-com bubble	Software	3	17.6
Dot-com bubble	Internet Retail	2	11.8
Dot-com bubble	Communications Equipment	2	11.8
Dot-com bubble	Semiconductors	2	11.8
Dot-com control	Semiconductors	4	13.3
Dot-com control	Semiconductor Equipment	4	13.3
Dot-com control	Software	3	10.0
Dot-com control	Restaurants / Consumer Services	2	6.7
AI-positioned	Technology	24	40.0
AI-positioned	Healthcare	7	11.7
AI-positioned	Financials	6	10.0
AI-positioned	Consumer Goods	5	8.3
Non-AI controls	Technology	15	35.7
Non-AI controls	Financials	5	11.9
Non-AI controls	Retail	5	11.9
Non-AI controls	Industrials	3	7.1

4.2.2 Price-to-Sales Comparison

Table 4.4 reports the mean and median P/S ratios for the dot-com and AI-era firm groups, distinguishing between bubble or AI-positioned firms and their respective control samples. The results show a clear difference between the dot-com bubble firms and the AI-era firms. In median terms, dot-com bubble firms display much higher P/S ratios, ranging from 17.48 in 1999 to 22.57 at the dot-com peak, while AI-positioned firms range only from 4.16 to 4.79 across the AI snapshots. This suggests that the typical AI-positioned firm is not valued at revenue multiples comparable to the typical dot-com bubble firm.

A key feature of the table is the large gap between mean and median values, especially in both control groups. This indicates that the distributions are strongly right-skewed, meaning that a limited number of firms with extremely high P/S ratios pull the mean upward. As a result, the mean reflects the influence of extreme upper-tail observations, while the median provides a more reliable measure of the central valuation level of the typical firm, and relying only on mean values could overstate the valuation level of the broader group and create the impression of stronger bubble-like pricing than what is visible for the median firm.

Table 4.4: Mean and median P/S ratios across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms)

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	Affected firms		Control firms	
Group	Mean	Median	Mean	Median
Dot-com 1999	35.32	17.48	362.87	4.07
Dot-com 2000	29.68	19.40	197.19	5.20
Dot-com peak	38.93	22.57	325.90	5.32
AI 2024 max	6.58	4.39	103.17	3.49
AI 2025 max	6.62	4.79	122.73	4.16
AI 2026 max	5.76	4.27	107.90	3.98

In more detail, as also shown in Fig. 4.4, this pattern suggests that AI-positioned firms consistently have a P/S ratio above the non-AI control group, but both have low levels when compared with the much higher median valuations observed among dot-com bubble firms. At the same time, the AI-positioned firms remain clearly above the non-AI controls across all AI snapshots, indicating that the market adds extra valuation to firms considered more strongly positioned in AI.

The figure excludes outliers according to the IQR rule described earlier in section 3.4.2, so the visualization focuses on the main body of the distribution rather than the most extreme observations.

A further point of interest is that both AI-positioned firms and non-AI controls remain much closer to the dot-com control group than to the dot-com bubble group. This suggests that, although AI-related hype is visible in the sample, the overall level of P/S valuation in the AI era is still more comparable to historically non-bubble firms than to the more extreme pricing observed at the dot-com peak. To confirm that this pattern is not driven by differences in terms of sector/industry or maturity of the involved firms between the groups, the P/S comparison is repeated while controlling for these two characteristics in Appendix A.9, A.10 and A.11 figures, where the same pattern is observed.

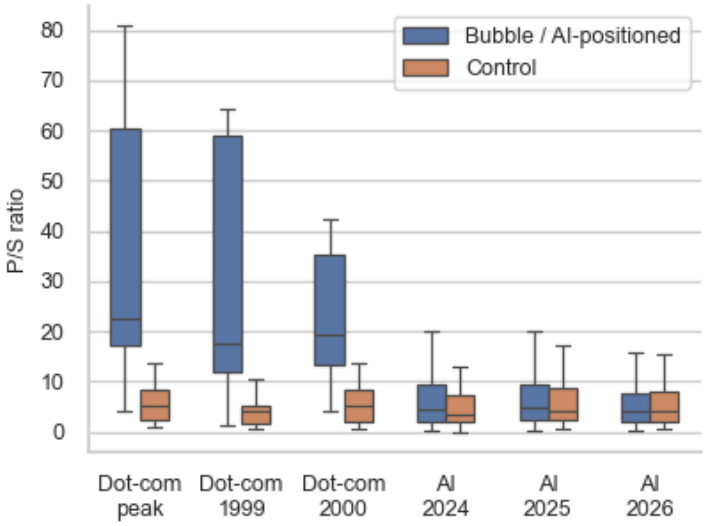


Figure 4.4: Distribution of P/S ratios across dot-com and AI firm groups.

4.2.3 Price-to-Earnings Comparison

Table 4.5 reports the mean and median P/E ratios for the dot-com and AI-era firm groups, separating the bubble or AI-positioned firms from their corresponding control samples. A notable feature of the table is the large gap between mean and median values, especially in the dot-com groups and the AI control group. This difference indicates that the P/E distributions are strongly right-skewed: a small number of firms with extremely high P/E ratios pull the mean upward, while the median better reflects the valuation level of the typical firm. Therefore, just like the P/S analysis, the median is more informative for comparing central valuation levels across groups, whereas the mean highlights the presence of extreme observations and speculative pricing outliers.

Figure 4.5 shows a spread in P/E ratios across all groups, with the dot-com bubble samples displaying particularly extreme high observations. In median terms, the dot-com bubble group ranges from 32 to 130 at the dot-com peak, while the dot-com control group ranges from 37 to 50 at the peak. In the AI-era sample, the median P/E ratio for AI-positioned firms ranges between 25.40 and 28.71, while the non-AI control group ranges between 24.92 and 31.88.

Table 4.5: Mean and median P/E ratios across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms).

	Affected firms		Control firms	
Group	Mean	Median	Mean	Median
Dot-com 1999	663.68	95.29	3342.59	37.05
Dot-com 2000	161.04	32.32	2642.60	45.08
Dot-com peak	651.55	129.33	3002.28	52.91
AI 2024 max	38.79	28.71	499.27	24.92
AI 2025 max	99.60	28.42	646.20	31.88
AI 2026 max	33.26	25.40	592.09	25.73

This pattern suggests that AI-positioned firms are not priced at the same earnings multiple levels as the dot-com bubble firms. Instead, both AI-positioned firms and non-AI controls appear much closer to the dot-com control sample than to the dot-com bubble sample. At the same time, the difference between AI-positioned firms and non-AI controls is narrower for P/E than it was for P/S, which indicates that the AI-positioned firm is less pronounced when valuation is measured relative to earnings rather than revenue.

A further point of interest is the movement within the AI sample over time. The median P/E ratio for AI-positioned firms is 28.71 in the 2024 maximum snapshot, 28.42 in 2025, 25.40 in 2026, and 25.70 in the April 2026 last observation. For non-AI controls, the corresponding medians are 24.92, 31.88, 25.73, and 26.69. This suggests that the AI and non-AI groups follow similar levels of earnings-based valuation over the observed period, with no persistent widening of the AI-positioned firms.

Also, in Fig. 4.5, it is shown that the dot-com groups, especially the bubble ones, have a wider spread and more extreme observations, while the AI-era groups cluster more closely together in their central range.

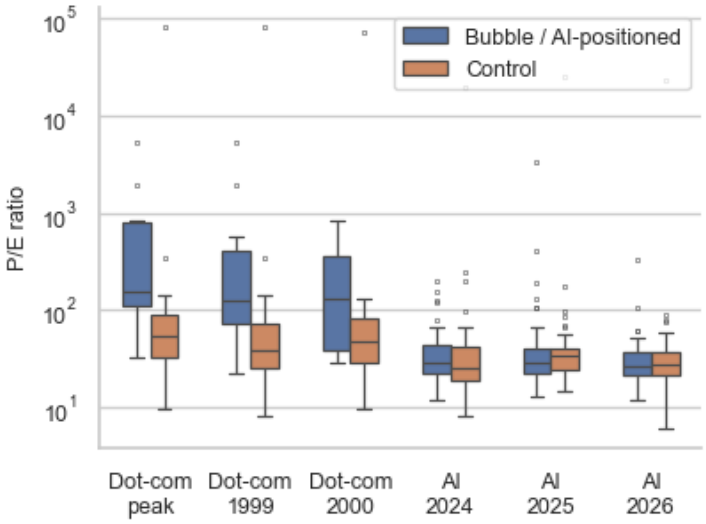


Figure 4.5: Distribution of P/E ratios across dot-com and AI firm groups.

Overall, the P/E results suggest that the AI-era sample does not replicate the extreme earnings-based valuation levels associated with the dot-com bubble. Although AI-positioned firms have a bit higher value than non-AI controls, the difference is limited and not consistently widening over time. In this respect, the P/E comparison points to a more stable pattern and indicates that AI-positioned firms' valuation is less dramatic when assessed relative to earnings.

4.2.4 EV/EBITDA Comparison

The EV/EBITDA is available as a single annual value in 1999 for the Dot-com groups and in 2025 for the AI era groups. The summary statistics are reported in Appendix Table B.2 and they show that all groups have very similar mean and median values, except for the dot-com bubble group, which contains much larger upper-tail observations that strongly affect the mean.

Figure 4.6 shows that the median EV/EBITDA ratio is 19.23 for dot-com bubble firms and 26.82 for dot-com control firms. In the AI-era sample, the median is 16.91 for AI-positioned firms and 17.74 for non-AI controls. This indicates that the AI-positioned and non-AI groups are very close to one another on this metric, with only a limited difference in central tendency. It also suggests that the AI-era medians are below the dot-com control group and broadly similar to the dot-com bubble median, although the dot-com bubble sample shows much greater spreading.

Figure 4.6 also shows that the EV/EBITDA distributions are spread out in the dot-com sample. Several large outliers were observed in the dot-com bubble sample, but far fewer in the AI-era samples. These extreme values strongly affect the means, but the medians remain much more stable and therefore provide a more reliable basis for comparison. The EV/EBITDA comparison suggests that the difference between AI-positioned firms and non-AI controls appears very small compared to their P/S and P/E levels difference, which implies that once valuation is scaled by operating earnings, the distinction between the two AI-era groups becomes less noticeable.

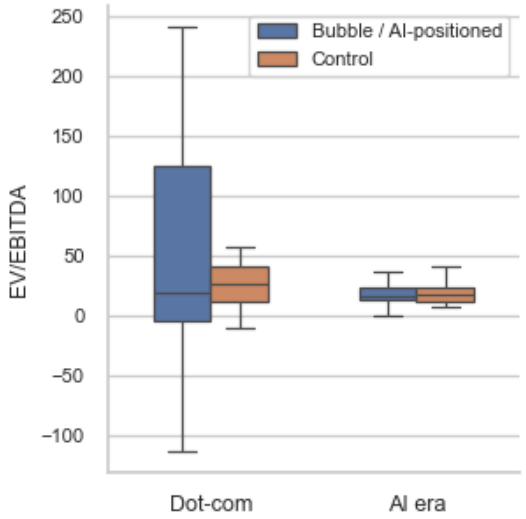


Figure 4.6: Distribution of EV/EBITDA multiples across dot-com and AI firm groups.

4.2.5 EBITDA Margin Comparison

Figure 4.7 compares the distribution of EBITDA margins across the dot-com and AI-era groups. The summary statistics are reported in Appendix Table B.3; these show that AI-positioned firms record the highest median EBITDA margin at 26.10, followed by non-AI controls at 22.95, while the dot-com control and bubble groups record lower medians of 13.69 and 9.86, respectively. This suggests that AI-era firms, both AI-positioned and control groups, display stronger operating profitability than the firms in the dot-com sample.

The comparison also shows that the distribution of profitability is highly uneven, especially in the dot-com bubble group. Several firms in this group have very large negative EBITDA margins, which strongly lowers the mean to -148.84 despite a positive median of 9.86. A similar, though less severe, effect is visible among non-AI controls, where negative outliers reduce the mean to -17.09 even though the median remains positive at 22.95. This reinforces the importance of using the median when comparing typical profitability across groups, since the mean is highly sensitive to extreme loss-making observations.

Overall, the EBITDA margin comparison suggests that AI-era firms are not only valued differently from the dot-com groups but also appear operationally stronger. In contrast to the dot-com bubble sample, where profitability is weaker and extreme negative margins are more common, AI-positioned firms show higher typical margins and less severe downside in the central distribution. This indicates that the AI-era sample is supported by stronger operating profitability than the dot-com bubble sample.

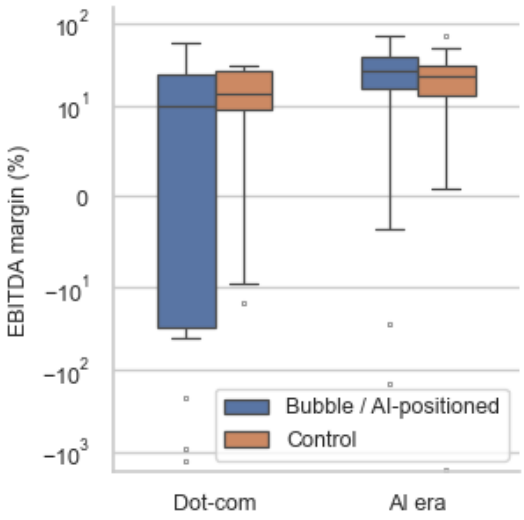


Figure 4.7: Distribution of EBITDA margins across dot-com and AI firm groups.

4.2.6 Recovery Perspectives

In addition to comparing valuation and profitability measures, the recovery perspective provides a way to assess how long firms took to regain value after the dot-com period. This adds a post-bubble dimension to the firm-level analysis and helps distinguish between firms that were able to recover relatively quickly and those that remained impaired for longer periods.

The descriptive results in Table 4.6 show a clear difference between dot-com bubble firms and dot-com control firms. Dot-com bubble firms required, on average, 13.62 years to recover, with a median recovery time of 14 years. By contrast, the dot-com control group recovered substantially faster, with an average of 7.96 years and a median of only 5 years. This indicates that the control firms in the Dot-com sample were generally able to return to prior valuation levels more quickly than the bubble firms. The distributional pattern is also shown visually in Appendix Figure A.7, where delayed recovery is more common among the dot-com bubble firms.

This difference in recovery speed suggests that firms associated with stronger bubble dynamics not only exhibited more stretched valuations but also faced a longer path back to recovery after the market correction. In that sense, the recovery comparison complements the valuation evidence by showing that the distinction between bubble and control firms was not limited to financial multiples, but also extended to the duration of post-bubble adjustment.

Table 4.6: Recovery years across dot-com bubble and dot-com control firms

Group	Mean	Median
Dot-com bubble	13.62	14.00
Dot-com control	7.96	5.00

The recovery perspective is limited to the dot-com sample in the present analysis, but it still provides an informative benchmark for interpreting the possible longer-term implications of bubble-like valuation patterns.

5. Discussion

In the previous chapter, we presented results related to the similarity between the dot-com and AI eras. This is done using the most important multiples and the most representative indices. More specifically, this chapter provides: a) a comprehensive discussion of the most important findings; b) how these findings relate to the research questions of this thesis and to the existing literature; and c) the limitations and possible future extensions of the research of this thesis. Its objective is to highlight clearly the main contributions of this thesis with respect to the RQs and hypotheses. As the findings of this thesis are mixed, both the evidence that the AI era is not a simple repetition of the dot-com bubble and the alarming signals that indicate bubble formation during this era are elaborately discussed. In the next step, the interpretation of the findings of this thesis is done in relation to its RQ and prior work. Finally, the limitations of the current work are discussed.

5.1 Key Findings

The main finding obtained from the results presented in Chapter 4 shows a mixed situation. In more detail, it can be seen that the AI era exhibits similarities with the dot-com era, up to a level, but there are also important differences. This is the case both for the market and firm level analyses.

At the market level, as inferred by the analysis of the NASDAQ Composite, NASDAQ 100, and SP 500 indices in Section 4.1, the AI era resembles partly the dot-com acceleration phase, but it cannot be seen as a repetition of the dot-com era. Despite that, both eras share some similarities in terms of direction and in terms of the early stages of a bubble cycle, as discussed in Section 2.1.3, the similarity measures of correlation and DTW distance on the trends in Table 4.1 show only moderate similarity. In Table 4.1, the values of correlation range from 0.59 to 0.63 while the (normalized) DTW distance ranges between 0.4 and 0.49.

Furthermore, the evolution of the normalized indices in Fig. 4.1 shows that the AI era follows a more gradual and temporary spread upward evolution movement which is interrupted temporarily by a number of market corrections and recoveries. Note that these interruptions are generated not by the sentiment of the market but from exogenous events, such as the war in Ukraine and the rhetoric of trade wars of President Trump. In contrast, during the dot-com era and for the days between 1000 and 2000, the indices (especially NASDAQ Composite and NASDAQ 100) experience explosive and sharp upward and then downward movements. The previous observations suggest that the AI era exhibits a more sustainable evolution. Note that the previous inference is also supported by systemic reasoning, which justifies the upward direction of the market during the AI era (persistent low rates, especially during 2015- 2021; large monetary expansion; passive investing and index funds; corporate buybacks; inflation and nominal growth; governmental support after Covid).

Another inference that can be drawn from the evolution of the indices is connected with the market cycle framework of a bubble (Kindleberger, 1978). The dot-com era clearly follows a complete bubble cycle, from the early adoption phase to the consolidation phase, as predicted by Kindleberger (1978). In contrast, there is strong evidence that the AI has not completed yet a bubble cycle but it is still in an ongoing combined early adoption and acceleration phase. This is supported by the rolling DTW distance results between the AI era and the different phases of the dot-com era, presented in Fig. 4.2. More specifically, based on the rolling DTW distance results, we can see that the AI era exhibits high similarity with the early adoption phase of the dot-com era from 2018 up to 2020. Furthermore, the AI exhibits similarity with the acceleration phase in two periods, between 2021 and 2023, and between 2024 and 2025. As there is no continuity between the two phases, but they are limited to isolated periods, it can be concluded that the AI is still in an ongoing market cycle development. However, it is worth noting that based on the rolling DTW distance results, since 2025 and up to the moment of writing this thesis (2026), the AI exhibits similarities with the mania phase of the dot-com era, which is an alarming signal.

Focusing on specific indices, we can see that the NASDAQ 100 index during the AI shows reduced similarity with the phases of the cycle of the dot-com bubble, especially with the mania and the crash phase, despite being a heavy technology index. This is the result of the different evolution of the large

cap firms that constitute NASDAQ 100 in the two eras. This suggests that the technology leaders of the AI era are less speculative than the technology leaders of the dot-com era. In addition, the SP 500 index exhibits relatively high similarity both in terms of rolling DTW distance but also in terms of correlation and DTW distance. This suggests that the dynamics of the AI boom are spread in the general market, creating a larger diffusion compared to the dot-com era, as described by Rogers (2003).

In terms of volatility-return connections, it is found that the two eras share the same central behavior but different risk-return distributions. In more details, both eras exhibit central behavior of low-to-moderate volatility and on average almost zero returns. However, the dot-com bubble appears to have a wider and more continuous volatility, which aligns with the stylized facts of asset returns from empirical finance (Cont, 2001). Whereas, the AI era appears to have a narrow volatility combined with islands of high volatility. From the previous analysis, it is inferred that the dot-com era was in a high risk regime for extended time periods, while, the AI experienced spikes of high risk regimes mainly due to external events.

The firm-level results also support the position that there are differences between the two eras. The P/S and P/E results show that the AI-positioned firms did not reach the same extreme valuation levels as dot-com bubble firms. This indicates a distance from the bubble creation for the current stage of the AI era according to the relative decoupling and fundamental divergence framework (Campbell et al., 1997a; Shiller, 1981; West, 1987) due to the absence of disconnection between pricing and fundamentals of a firm. AI-positioned firms were sometimes valued higher than non-AI controls, but the difference was not large. In several comparisons, both AI groups were closer to the dot-com control firms than to the dot-com bubble firms. The profitability results support the same interpretation. The EV/EBITDA results showed small differences between AI-positioned firms and non-AI controls. This means that the difference became smaller when valuation was compared with operating earnings. The EBITDA margin results also showed that AI-era firms had stronger profitability than the dot-com firms. This suggests that the AI boom may include high expectations, but these expectations are linked to firms with stronger fundamentals.

The recovery results add a historical warning. Dot-com control firms recovered faster than dot-com bubble firms after the crash. This suggests that firms with less extreme valuations were more resilient. The same recovery analysis cannot yet be applied to the AI era, because the cycle is still ongoing. Still, the dot-com recovery pattern is useful when thinking about the risk of stretched valuations.

Overall, the main finding is that the AI boom shows some bubble-like signals, but it does not directly repeat the dot-com bubble. The current period appears more supported by firms fundamentals. It is also less extreme in both market-level patterns and firm-level valuation multiples. The following sections discuss this interpretation in more details.

5.2 Evidence Against a Direct Dot-Com Bubble Repetition

The most clear result obtained from the analysis of this thesis is that the AI boom does not appear to be a direct repetition of the dot-com bubble. Some similarities exist, but there are also significant differences. These differences are clearly visible in the level of valuation multiples and the evolution of the most representative indices.

5.2.1 Stronger Fundamentals in the AI-Positioned Firms

One important difference is that many firms in the AI era are more established than the firms associated with the dot-com bubble. As shown in Table 4.2, the AI-era groups include more mature firms than the dot-com bubble group. The sample of AI-positioned firms also includes several firms with positive revenues and profits and strong market positions. This is important because bubbles are often understood as cases where prices move beyond what can be explained by fundamentals (Lansing, 2008). The AI-era sample, therefore, differs from many dot-com firms, where valuations were often based more on future expectations than on current operating strength.

The valuation multiples also support this interpretation. The P/S comparison in Fig. 4.4 shows that AI-positioned firms have slightly higher values than non-AI controls in some periods, but they are still far below the dot-com bubble firms. A similar pattern is visible in the P/E comparison in Fig. 4.5. This

suggests that AI valuations may be somewhat elevated, but they do not reach the extreme levels seen during the dot-com bubble.

The profitability results support this difference. The EBITDA margin comparison in Fig. 4.7 shows that the AI-era firms have stronger operating profitability than the dot-com firms, especially compared with the dot-com bubble group. This means that higher expectations around AI are not only linked to young or unproven firms. They are also linked to firms that already generate operating earnings. This matters because EBITDA based measures connect valuation to operating performance, which makes them useful when assessing whether market prices are supported by fundamentals (Mauboussin and Callahan, 2024b; Nenkov, 2025). The difference is also visible in the EV/EBITDA comparison in Fig. 4.6, where the AI-era groups appear less extreme when valuation is compared with operating earnings. Based on the previous observations and on the relative decoupling and fundamental divergence framework (Campbell et al., 1997b; Shiller, 1981; West, 1987), the AI cannot be characterized as a bubble yet.

5.2.2 Broader AI Adoption Across Sectors

Currently, AI is either integrated or in the process of being integrated in firms across different sectors. This is consistent with the view of AI as a GPTech rather than only a narrow speculative theme (Brynjolfsson et al., 2021; OECD, 2025b). This matters because the AI boom is not limited to young technology firms with uncertain business models. It is also connected to established firms in sectors such as retail, healthcare, finance, and insurance, etc. In many cases, non-technology firms are not only simple users of AI tools, but also important drivers of AI adoption and development. This can be seen in recent external AI adoption trackers. The Salt Technologies Fortune 500 AI Adoption Tracker (Salt Technologies AI, 2026) lists several non-technology firms as AI maturity leaders, including Walmart in retail, JPMorgan Chase, Visa, and Mastercard in financial services, and Coca-Cola in consumer goods.[1] Similarly, the Larridin AI Tracker (Larridin, 2026) assigns the highest possible AI score in its scale to several firms outside the traditional technology sector, including Bank of America and BlackRock in financials, Eli Lilly and GE HealthCare in healthcare, and DoorDash in consumer discretionary.[2]

This broader adoption pattern makes the current AI period different from a narrow speculative cycle. The interest in AI is not only concentrated in a small group of pure technology firms. It is also connected to practical adoption across established sectors. This suggests that part of the market interest in AI is linked to real business use rather than only speculative expectations. Such a suggestion indicates a narrower diffusion-evaluation gap (Brynjolfsson et al., 2017; David, 1990) in the sense that the adoption of AI and the generation of profits from it has a smaller time offset.

5.2.3 Limited Separation Between AI-Positioned Firms and Controls

Another important result is the limited separation between AI-positioned firms and non-AI control firms in terms of valuation. In general, firms with stronger AI positioning are expected to receive a premium valuation because investors expect higher future growth (Watanabe et al., 2022). If the current period reflected a broad AI bubble, this premium would be expected to appear as a large and consistent gap between AI-positioned firms and non-AI controls according to the relative decoupling and fundamental divergence framework (Campbell et al., 1997b; Shiller, 1981; West, 1987).

The results do not show such a strong separation. In the P/S comparison in Fig. 4.4, AI-positioned firms were valued somewhat higher than non-AI controls, but both groups remained on similar levels and far below the dot-com bubble firms. The P/E comparison in Fig. 4.5 showed an even closer pattern, where the AI-positioned and control groups moved at similar levels across the observed period. The same point is supported by Fig. 4.6 and Fig. 4.6 when valuation was compared with operating earnings, the difference between AI-positioned firms and non-AI controls became very small.

This weakens the argument that the market is applying an extreme AI-specific premium across the full sample. It suggests that AI exposure may matter, but it does not appear to create the kind of broad valuation separation that would be expected in a dot-com style bubble.

[1] Salt Technologies AI, *Fortune 500 AI Adoption Tracker 2026*: `https://www.salttechno.ai/datasets/fortune-500-ai-adoption-2026/`.

[2] Larridin, *AI Adoption Tracker*: `https://data.larridin.com/data/ai-tracker/`.

5.2.4 A More Sustainable Market Evolution

As shown in Section 4.1 and discussed in Section 5.1, despite the similarities between the two eras, the evolution of the normalized indices of NASDAQ Composite, NASDAQ 100, and SP 500 clearly shows that there are also important differences. In particular, these differences indicate that the AI era is more sustainable, up to the point of writing of this thesis, because: a) its evolution is spread wider in time and more gradual; b) it is less speculative due to lower volatility; and c) it is more connected to the operating fundamentals.

In more details, up to the point of writing of this thesis, the technology indices of NASDAQ Composite and NASDAQ 100 and partly the general market index of SP 500 show a constant gradual upward evolution, which in certain points in time and due to mainly external factors (war in Ukraine, Covid, etc.), is is interrupted by down-ward movements and renewed advances. This is also supported by the rolling DTW distance analysis in Section 4.2, which shows that the evolution of the indices during the AI era is in an ongoing early adoption and acceleration phase up until late 2025 to late 2026, where the first elements of the mania phase appear. Note that, as there is a short period since the appearance of these mania elements, it is difficult to conclude whether we are in the mania phase. Thus, it is clear that the AI era has not completed yet the bubble cycle predicted by (Kindleberger and Aliber, 2005) and the Kindleberger–Minsky theoretical framework.

More indications regarding the more sustainable evolution of the AI era compared to the dot-com bubble can be found in the analysis of the relation between volatility and return given in Section 4.1.4. The AI era despite having islands of very high volatility, these islands are time limited, while, the total volatility-return distribution has a very central nature. More specifically, the central region of the AI era is strongly concentrated in mid-low to moderate volatility and approximately zero return. In contrast, the dot-com era, even though it has a central volatility-return distribution, the centrality is significantly weaker than the AI era, while, its volatility-returns distribution extends continuously and with higher density to higher levels than the main region of the corresponding distribution of the AI era. The previous behavior is in strong connection with the stylized facts from empirical finance which predict volatility clustering, fat tails, and asymmetric draw-downs, etc. (Cont, 2001). Another connection can be found with the Phillips–Yu tradition (explosive dynamics and rational bubble framework), which identifies bubbles from their explosive phases of price change (Phillips et al., 2015).

Furthermore, the growth during the AI era is not limited only to the technology indices (NASDAQ Composite and NASDAQ 100) but also to indices that capture the general market, such as SP 500. This supports the claim that AI is a GPTech, not a narrow speculative phenomenon of a specific technology sector. Combining this conclusion with the finding of Section 5.2.3, where it is discussed that the valuation of AI positioned firms does not differ a lot with the valuation of non AI positioned firm, it can also be inferred that the evolution of the indices during the AI ere is also support by the fundamental valuation.

5.3 Bubble Signals in the AI Era

Although the overall evidence does not support a direct repetition of the dot-com bubble, some results still point to bubble-like risks. These signals are not strong enough to define the whole AI era as a bubble, but they are important as they might indicate a bubble in the process of formation. They are mainly related to market concentration, temporary similarity to the dot-com acceleration phase, and strong investor expectations around AI.

5.3.1 Market Structure, Breadth, and Volatility

As already discussed, there are signals connected with the structure, breadth, and volatility of the market that indicate either the formation of an AI bubble or even an already AI bubble. Firstly, in terms of structure, we can see from Fig. 4.2c that, since the middle of 2025, the resemblance of the AI era with the mania phase of the dot-com era is in general increasing while the resemblance with the acceleration phase is consistently decreasing. This might be an indication of a market transformation towards a less sustainable upward movement if the previous pattern is proven strong in time. Furthermore, in terms of structure, the high capex of the AI infrastructure is a characteristic connected with increased risk. If the cash generation is delayed or has a weaker increasing monotonicity compared to the capex, the formation of a bubble will become more probable. Finally, another alarming structural signal comes from the high

market concatenation which according to (Mauboussin and Callahan, 2024a) is reaching historical highs. A strong example of this is capitalization of the Magnificent Seven firms, i.e. Apple, Microsoft, Alphabet, Amazon, Nvidia, Meta Platforms, and Tesla, which in 2025 had a combined capitalization of $22 trillion dollars which constituted approximately the 33% of SP 500 (Brock, 2026), while now it is reaching to approximately 50% (Kokkinidis, 2025).

A characteristic of the AI era is the narrow breadth signal, which shows that despite the large gains and the evolution of the SP 500, explosive gains are mainly obtained by a number of AI and technology heavy firms, such as the Magnificent Seven (Mauboussin and Callahan, 2024a). In particular, the 33% exposure of SP 500 to the Magnificent Seven, as given in (Brock, 2026), may limit the wider diffusion of the AI dynamics to the general market as captured in Section 4.1. Another alarming signal comes from the passive investments, i.e., index funds or Exchange Traded Funds (ETFs), and the way they are directed. As the capitalization of large AI firms increases in the technology or most tracked indices, more index funds or ETFs allocate more investment to these large AI firms. This may push prices to a higher level via a loop of index weight, direction of investments of funds, and valuation. This is consistent with the concentration-driven bubble dynamics and self-reinforcing market patterns theoretical frameworks described in Section 2.4.

Lastly, in terms of volatility, Fig. 4.3 clearly shows that the AI era exhibits islands of very high volatility, especially to the technology heavy indices of NASDAQ Composite and NASDAQ 100, despite the high centrality of the volatility-returns distribution. As already described in Section 4.1.4, these islands can be explained by exogenous events that coincided in time (War in Ukraine, trade war rhetoric of President Trump, etc). However, there might be an indication of moving towards a more unstable regime or that they are the result of sudden shifts in AI expectations, macro shocks, or technology-sector repricing. In addition, the very high volatility islands of the AI era might be an indication of fat-tail behavior. This interpretation is motivated by the stylized facts of financial markets (Cont, 2001), including fat tails, volatility clustering, and asymmetric drawdowns. The fat-tail interpretation is important as it establishes that large market movements may occur even when the volatility regime appears stable. Another observation from Fig. 4.3 is that the SP 500 index has weaker islands of very high volatility compared to the technology indices of NASDAQ Composite and NASDAQ 100. This might be an indication that the technology sector is in a more unstable regime, which could potentially be interpreted as a technology-specific bubble-risk signal. In addition, the sharp short volatility islands during the AI era may endanger that heavy exposed investors to AI to proceed to sell-offs for risk re-balancing. If this is combined with other external events, such as higher interest rates or lower growth, the sentiment of the market may change for more prolonged periods, causing herding sell-offs. Such behavior is explained by the investor sentiment and herding mechanisms discussed in Section 2.4.

5.3.2 Investor Expectations Around AI

The expectations of investors are one of the reasons why the AI era can still show more bubble-like signals. Prior research shows that investor sentiment can move prices when beliefs about future cash flows are not fully supported by current facts (Baker and Wurgler, 2007). This risk is especially relevant in technology cycles, where investors may overreact to expected productivity gains from new technologies (Chang et al., 2016). Similar patterns were observed during the dot-com period, where technology stocks reached very high valuation levels and traditional accounting indicators became less connected to market prices (Morris and Alam, 2012).

The AI-era results are partly aligned with the expectation that AI is in the process of creating growth, lower costs, and new business opportunities. This can slightly raise the valuation of firms perceived as AI-positioned. This is partly visible in the P/S comparison, where AI-positioned firms were sometimes valued above non-AI controls in Figure 4.4. However, the difference is not large enough to resemble the extreme valuation levels of the dot-com bubble. Therefore, investor expectations around AI should be interpreted as a limited bubble-like signal, not as proof of a real bubble.

5.4 Interpretation in Relation to the Research Questions

The main RQ aims to answer to what extent the current AI hype resembles the dot-com bubble in terms of market-level cycle evolution and firm-level valuation patterns. Based on the results of this study, the

answer is that the AI boom shows partial similarity, but not direct repetition. The market-level results showed some similarity to parts of the dot-com cycle, especially to the early adoption and acceleration phases. However, the overall pattern was more gradual and less extreme than the dot-com bubble. The firm-level results also showed clear differences, since AI-positioned firms did not reach the same extreme valuations as dot-com bubble firms.

For the first sub-question, the market-level evidence suggests that AI-era indices show a non-negligible level of similarity to the dot-com period, but only in specific phases and time windows. Therefore, the results support the idea of partial similarity and not a complete cycle repetition. This means that **H1** is only partly supported. Also, the evidence does not support the stronger claim that the observed levels of similarity represent an identical repetition of the dot-com bubble. Similar conclusions are drawn from the distribution of the volatility and returns of the considered indices. Despite some similarities in terms of centrality, the distribution of the volatility and returns of the considered indices is different for the two eras.

For the second sub-question, the firm-level evidence showed that AI-positioned firms were sometimes valued higher than non-AI controls, especially in the P/S comparison. This provides some support for **H2**. However, the difference between AI-positioned firms and non-AI controls was very small. Therefore, H2 is supported only in a moderate way, rather than as evidence of a strong AI-specific valuation bubble.

Hypothesis **H3** is not supported by the results. The valuation patterns of AI-positioned firms did not mirror the dot-com peak and remained far below the dot-com bubble firms. This suggests that the current AI-positioned firms' valuation pattern is different from the dot-com peak, and **H3** is rejected.

Hypothesis **H4** is also rejected by the results. The hypothesis expected AI-era firms to show similar operating profitability to dot-com era firms. However, the EV/EBITDA and the EBITDA margin comparisons showed that AI-era firms had stronger, more stable operating profitability and appeared more financially mature compared with the dot-com bubble group.

Overall, the findings suggest that the AI boom includes some speculative and expectation-driven elements, but the broader evidence does not support calling it a direct repetition of the dot-com bubble. The current period appears more supported by mature firms, stronger profitability, and wider cross-sector adoption. The most balanced interpretation is therefore that the AI boom contains limited bubble-like signals, but it is not yet a dot-com style bubble based on the evidence in this thesis.

5.5 Findings in Relation to Previous Research

In this section, the alignments and deviations of the findings of this thesis with respect to prior work are explored. This is done for both the index and the firm-level analyses. This information is important because it places this thesis within the academic and market discussion on whether AI is a bubble.

Focusing on the index-level analysis, this thesis has shown that there are partial similarities between the two eras, but the AI era is not a simple replication of the dot-com bubble. This is clearly captured by both the visual analysis of Section 4.1.1, the similarity measurements of Sections 4.1.2 and 4.1.4, and the analysis of the distribution of the return-volatility relation of Section 4. The previous observations are consistent with (Perez, 2002), which claims that phases of financial excitement are generated by technological evolution across different economic and institutional environments. Due to differences in environments, the two financial excitement phases are not expected to be identical. A similar position is given by (Kindleberger and Aliber, 2005) where bubble cycles are formed from consecutive phases that do not share the same characteristics at all times. Finally, the same position is stated explicitly in (Letzing and Sung, 2025). Note that positions expressed in this thesis regarding the difference between the two eras are also in alignment with (Wattret, 2025).

An example of a difference between the two eras is the more sustainable market evolution (in connection with the financial fundamentals) during the AI era and the absence of explosive patterns of movements like the ones observed during the dot-com era and documented by (Ofek and Richardson, 2003). This observation is fully aligned with (Peterson, 2025), which claims that the main AI-positioned firms exhibit price evolution supported by stronger profits, cash flows, and balance sheets, rather than the speculative nature of the AI bubble.

As discussed in Section 4.1.2 and can be seen in Fig. 4.2, the results produced using the rolling DTW show that the AI era, up to the moment of writing, resembles the most with the early adoption and acceleration phases of the dot-com era rather than a complete bubble cycle. This is despite the longer

time spread compared to the dot-com era. As AI is framed as a GPTech (Brynjolfsson et al., 2021; OECD, 2025b), the previous observation is in alignment with the literature that predicts that GPTechs are slower in producing their full economic impact and growth (Bresnahan and Trajtenberg, 1992). This is due to the gradual adoption across sectors (Helpman and Trajtenberg, 1996) and the requirements of complementary investments whose productivity effect appears in a later stage (Brynjolfsson et al., 2021). The categorization of AI as a GPTech is also supported by the evolution of the SP 500 index.

In Section 4.1.4, it is shown that the AI era includes islands of very high volatility combined with lower levels of central volatility compared to the dot-com era. In the previous sections, the very high volatility of the AI era is attributed not to consistent systemic reasons but to coincidental events and idiosyncratic risk. In (Fink et al., 2010), such observations of sharp, isolated incidents of very high volatility indicate an incomplete bubble cycle. This is in agreement with the findings of this thesis.

In terms of firm-level analysis, the results also align with prior research on technology valuation, AI-positioning, and speculative cycles, while also showing important differences from the dot-com benchmark. Prior studies argue that high-growth, technology-oriented firms are difficult to value with standard cash-flow models and are therefore often assessed through market multiples (Fernandez, 2023; Festel et al., 2013). During the dot-com period, valuation multiples moved by far above historical levels, and the link between prices and accounting fundamentals weakened (Lansing, 2008; Morris and Alam, 2012). The firm-level results of this thesis are consistent with that historical pattern at the dot-com peak, but they do not show the same extreme levels in the AI era. The median of P/S and P/E ratios for AI-positioned firms remains well below that observed from the dot-com bubble firms in Figs. 4.4 and 4.5, while both AI groups were generally closer to dot-com control firms than to dot-com bubble firms.

These study findings are also in line with recent discussions that distinguish between strong AI-related market performance and a full dot-com-style bubble in firm valuations (Letzing and Sung, 2025; Peterson, 2025), which confirms that the AI era may include elevated expectations, but the firm-level evidence in this thesis does not suggest a huge overpricing of the same magnitude as during the dot-com peak.

In addition, the literature on AI-positioned firms also suggests that firms with stronger AI exposure may receive a valuation premium because investors expect future growth, innovation opportunities, and capabilities (Babina et al., 2024; Moro-Visconti, 2024; Watanabe et al., 2022). This is closely related to the results obtained in Chapter 4, which provides partial support for this view. AI-positioned firms were sometimes valued above non-AI controls, especially in the P/S comparison in Figure 4.4. However, the gap between the two AI-era groups was limited and not consistently widening over time. The P/E and EV/EBITDA comparisons in Figs. 4.5 and 4.6 showed even smaller differences once valuation was related to earnings and operating performance. This suggests that the market may recognize AI exposure, but not as strongly as in the dot-com era. This limited difference is important because prior work treats valuation premia as a means of identifying strategic AI positioning in market data (Alzeghoul and Alsharari, 2025; Henriksson and Nyqvist, 2022; Pastor and Veronesi, 2009). This study's results, therefore, support the existence of some AI-related pricing effects, but not the kind of broad valuation divergence that would be expected if the current period were a dot-com-style bubble at the firm level.

The profitability results point in the same direction. Because accounting measures may undervalue the invisible assets in innovation-driven and cutting-edge technology firms (Lev and Gu, 2016; Pastor and Veronesi, 2009), the EBITDA margin and EV/EBITDA provide a useful check on whether prices are supported by operating performance (Mauboussin and Callahan, 2024b; Nenkov, 2025). Both AI-era groups showed more stable median profitability than dot-com bubble firms, and the gap between AI-positioned firms and controls narrowed when valuation was linked to operating earnings.

Overall, the firm-level evidence aligns with prior work on technology booms, where prices can run ahead of fundamentals (Lansing, 2008; Morris and Alam, 2012), but not as the dot-com peak extremes (Letzing and Sung, 2025; Peterson, 2025). The sample of AI-era firms was more mature and sector-diversified. Also, the dot-com bubble firms recovered more slowly than controls after the crash (Table 4.6), in line with post-peak adjustment in market-cycle theory (Britannica, 2026). Because the AI cycle remains incomplete, this recovery pattern is only a historical benchmark, although the lower current AI multiples still suggest a less extreme recovery period would be needed, compared to the dot-com era.

Finally, this thesis extends the previous research gap discussed in Section 2.5 by providing a more integrated empirical comparison between the dot-com and AI periods. Dobre et al. (2020) provide an early comparison of the two periods, but their sample ends in 2018 before the recent generative AI boom. Potrykus (2024) is partially close to this thesis because it uses DTW distance, but it mainly focuses on stock-price pattern similarity checks. Floridi (2024) and Cao (2025) provide theoretical explanations

of the AI boom, but they do not empirically combine market-cycle behavior with firm-level financial fundamentals. The findings of (Cao, 2025) are partly aligned with this thesis. However, the findings of (Floridi, 2024) are different than this thesis as it claims high similarity between the AI and dot-com era. Defend et al. (2026) focus mainly on portfolio characteristics, while Chelikavada and Bennett (2026) examine broader hype indicators such as scientific activity and market signals.

In contrast, this thesis combines market-level evidence, including index performance, returns, volatility, correlation, and DTW, with firm-level valuation and profitability measures such as P/S, P/E, EV/EBITDA, and EBITDA margin. Therefore, the findings suggest that the current AI boom shares some characteristics with the dot-com period, but it does not yet show the same extreme speculative pattern observed during the dot-com peak. Overall, this thesis contributes to the literature by demonstrating that the AI bubble question should be assessed through both market indecis and the firm-level financial fundamentals, and with a broader set of analysis tests and multiples.

5.6 Reflections and Limitations

The results provide useful insights, but they should be interpreted with some caution. The following limitations mainly concern the market-level method, the firm-level data, and the ongoing AI cycle.

5.6.1 Market-Level Methodological Limitations

The market-level analysis provides some insightful conclusions, but they should be treated with caution due to its methodological limitations. Firstly, the market level analysis is trying to find similarities between the AI and the dot-com era rather than categorize the AI era as a bubble or not. As there are structural differences between the two eras, especially in terms of concentration, the bubble creation during the AI may follow a different path. Furthermore, the indices are good indicators for the evolution of the market or of a specific sector, however, they fail to capture firm-level differences which might be important. The similarity metrics in this thesis are imperfect as they only capture pattern similarities in terms of shape (DTW distance and visual interpretation) and statistical behavior (correlation). Although the similarity results are interpreted in connection to their causal events, a more complete framework needs to consider the causal reasoning inherently. The same holds for the volatility-returns analysis of Section 4.1.4. The analysis of Section 4.1.4 captures the existence of the islands of high volatility during the AI, but it does not give strong justification of why such behavior arise. Furthermore, the AI era is still ongoing, thus some of the conclusions may need to be revised as time evolves. Therefore, the conclusions drawn from the market level analysis should be considered as warnings rather than definite conclusions.

5.6.2 Firm-Level Data and Sampling Limitations

The firm-level analysis also had several limitations. The dot-com and AI samples were not identical populations. They differed in firm age, sector composition, market context, and sample size. Therefore, the comparison should be read as a historical benchmark comparison, not as a direct one-to-one comparison between identical groups.

Another limitation was the availability of historical dot-com data. Many firms from the dot-com period no longer exist, and consistent financial multiple data were difficult to collect for these firms that failed after the crash. This creates survivorship bias because the dot-com sample mainly includes firms for which enough data could still be found. As a result, the dot-com benchmark may under-represent the most fragile firms from that period.

The financial measures were also not equally available across both eras. P/S and P/E were available across fewer time points during the dot-com era. In addition, the data included strong outliers, especially in the dot-com sample. Medians and outlier-aware figures were therefore used, but the interpretation still depends on the data's quality and completeness.

Finally, the classification of firms also depended on available external information. The Larridin (2026) AI ranking provides a useful and consistent basis for separating AI-positioned firms from controls, but it still reflects one external scoring approach. The dot-com classification was even less standardized, since it depends on historical sources such as archived Wikipedia articles, old online magazine articles, and archived web pages. Different classification methods could therefore produce slightly different samples and firm-level comparisons.

6. Conclusion and Future Research

6.1 Summary

This thesis examined whether the current AI hype resembles the dot-com bubble in terms of market-level indices evolution and firm-level valuation patterns. The current study used two complementary perspectives. The first perspective compared market indices from the dot-com and AI periods in order to examine broad market behavior. The second perspective compared firm-level valuation multiples and profitability indicators across the dot-com and AI eras, dot-com bubble and AI-positioned firms with corresponding control firms in each era.

The results showed that the AI boom has moderate similarities with the dot-com period, but the evidence does not support a direct repetition of the dot-com bubble. At the market level, the AI-era indices showed some similarity to parts of the dot-com cycle, especially the early adoption and acceleration phases. However, the overall market pattern was more gradual and less extreme than the dot-com bubble. At the firm level, AI-positioned firms did not reach the same valuation extremes as dot-com bubble firms. The results also showed that AI-era firms had stronger and more stable operating profitability than the dot-com bubble group.

6.2 Conclusion

The main research question of this thesis is formed as:

> To what extent does the current AI hype resemble the dot-com bubble in terms of market-level cycle behavior and firm-level valuation patterns?

The final answer to this question is that the current AI hype resembles the dot-com bubble only partially. The market-level analysis showed some similarities in specific phases and time periods, but not enough to describe the AI era as an identical repetition of the dot-com bubble. The firm-level analysis also weakened the bubble assumption, because AI-positioned firms were generally valued far below dot-com bubble firms and were closer to non-AI controls and dot-com control firms in several comparisons.

The first sub-question asked whether the AI-era market indices show similarities to the dot-com period. The results suggest that they do, but only in a limited way. The strongest similarity appeared around parts of the early adoption and acceleration phases, while the broader pattern remained less extreme than the dot-com price movements. Therefore, the market-level evidence supports partial similarity rather than a full bubble repetition.

The second sub-question asked how AI-positioned firms compare with non-AI controls and with dot-com firms. The results showed that AI-positioned firms sometimes had slightly higher valuation multiples than non-AI controls, especially in P/S. However, the difference was limited and did not consistently widen across the metrics. In addition, both AI-positioned firms and non-AI controls remained far below the valuation levels of dot-com bubble firms. Profitability indicators also showed that AI-era firms were more supported by operating fundamentals than many dot-com firms.

The hypotheses were therefore interpreted as follows. **H1** was partly supported, because some market-level similarity was found, but the evidence did not support an identical repetition of the dot-com bubble. **H2** was moderately supported, because AI-positioned firms sometimes showed higher valuation multiples than non-AI controls, but the difference was not large enough to indicate a broad AI-specific bubble. **H3** was rejected, because AI-positioned firms did not mirror the valuation patterns observed at the dot-com peak. **H4** was also rejected, because the hypothesis expected similar operating profitability, while the results showed stronger and more stable profitability among AI-era firms.

Overall, the thesis concludes that the AI boom contains some bubble-like signals, but it should not be interpreted as a direct repetition of the dot-com bubble based on the evidence in this study. The current period appears to be more supported by mature firms, stronger operating fundamentals, and broader cross-sector adoption of AI. At the same time, the presence of high investor expectations means that the risk of overvaluation cannot be dismissed.

Based on the previous discussion, the idea that AI is a temporal phenomenon with marginal effects is not supported, despite the moderate similarities of the AI era with the dot-com bubble. Throughout this thesis, but especially in Chapters 2, 4, and 5, AI is framed using both theoretical and empirical arguments (with strong preservation over time) as GPTech with large transformation strengths and extensive diffusion across different industries. This means that AI is becoming a fundamental element of the operation of firms, and their digital transformation and competitiveness. This indicates that AI is expected to shape the way business, science, etc, is done.

The conclusions should be interpreted with caution. The AI cycle is still ongoing, which means that a full rise, peak, crash, and recovery pattern cannot yet be observed. The study also relies on selected publicly listed firms and excludes private AI startups or venture capital valuations. In addition, the dot-com sample is affected by survivorship bias because many failed firms from that period no longer have complete financial data available. The analysis is descriptive and comparative. It does not estimate intrinsic firm value and does not establish a causal relationship between AI adoption and market valuation. The classification of AI-positioned and dot-com firms both depends on available external data.

6.3 Research Contributions

The main contribution of this thesis is that, until the moment of its writing, it is shown that the AI era is not the direct repetition of the dot-com bubble, despite the moderate similarities identified in this thesis. This is undertaken by combining market-level and firm-level evidence in one historical comparison. The market-level analysis provides a view of broad cycle behavior, while the firm-level analysis shows whether the pricing of individual AI-positioned firms resembles the dot-com period. This combination gives a more balanced view than using only index data or only firm-level multiples.

The thesis also contributes by comparing AI-positioned firms with non-AI controls rather than treating the entire current technology sector as one group. This distinction makes it possible to examine whether AI exposure is associated with clearly higher valuation multiples. The results suggest that AI positioning matters, but that the valuation gap is limited and does not resemble the extreme separation expected in a broad speculative bubble.

6.4 Future Work

Future research could extend this study in several ways. First, the analysis can be repeated when the AI cycle has developed further. A longer time period would make it possible to examine whether the current AI boom continues, corrects, or moves into a clearer recovery phase.

Second, future work could improve the firm-level sample. A larger dot-com sample, especially one that includes more failed or delisted firms, would reduce survivorship bias and provide a stronger historical benchmark. The AI-era sample could also be expanded by including more firms, more industries, and additional years of financial data. A richer sample would also allow more systematic assessment of how comparable the firm groups actually are, in terms of firm industry, sector, maturity, size, and operational characteristics. The comparability assessment in Appendices A.9 to A.11 is a small example in this direction, as it is limited to the industry type and firm maturity. Note that these characteristics are selected as they are the only consistently available across both eras.

Third, future studies could compare the AI boom with other historical market cycles. This could include technology-related periods, such as the Roaring Twenties during the 1920s, or non-technology bubbles, such as the housing bubble in 2008. This would show whether the AI era shares features with speculative market cycles different than what was observed in the dot-com bubble.

Fourth, this study has established that there are significant differences between the AI and dot-com eras. However, it provided only brief explanations of why such a difference exists. A future extension could consider systematically finding the reasons behind these differences.

Closing Remarks

Although the findings suggest that the AI boom does not directly repeat the dot-com bubble, the AI cycle is still ongoing. For this reason, the conclusions should be revisited as more data becomes available and as the long-term financial effects of AI adoption become clearer.

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A. Appendix: Figures

This appendix includes all the supporting figures referenced throughout the main text. Figures A.1 to A.5 provide the absolute, market-level evolution of the indices, while Figs. A.7 to A.11 support the firm-level comparability analyses.

A.1 Market-Level Index Figures

The following figures provide the absolute-value counterpart to the normalized index evolution discussed in Section 4.1.1. Because there is roughly a 10-year gap between the two eras, the absolute index levels differ substantially, so the main analysis relies on normalized values instead.

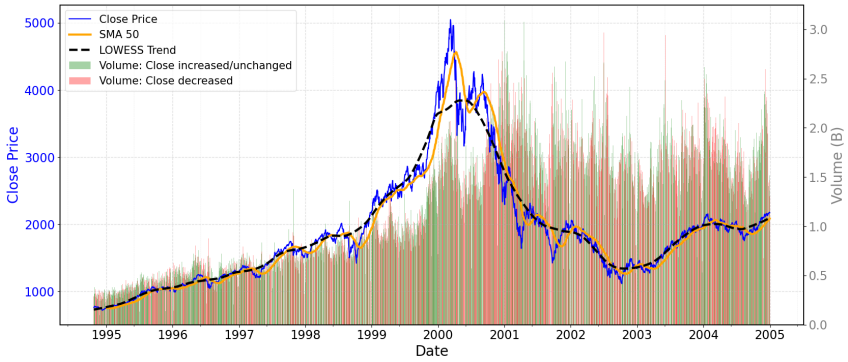


Figure A.1: NASDAQ Composite (Dot-com Era): Close, SMA(50), LOWESS , and volume

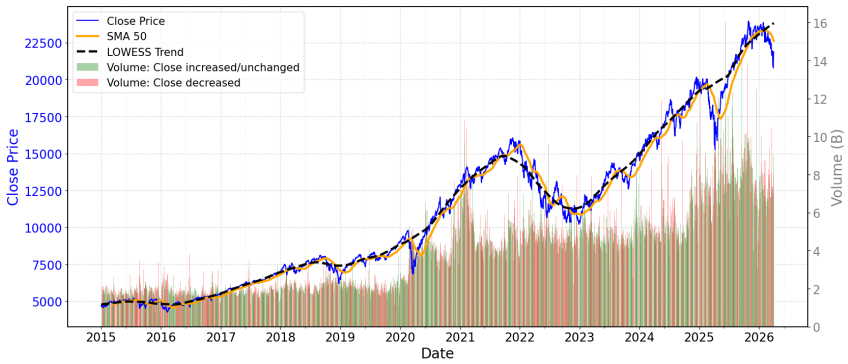


Figure A.2: NASDAQ Composite (AI Era): Close, SMA(50), LOWESS , and volume.

Figure A.1 shows the daily closing values of the NASDAQ Composite during the dot-com era, together with its 50-day SMA, a LOWESS trend, and the trading volume. The sharp run-up and subsequent collapse around the 1999–2000 peak are clearly visible. While, Fig. A.2 shows the same indicators for the NASDAQ Composite during the AI era. Compared with the dot-com era, the evolution is more gradual and spans a longer period, with corrections mainly associated with external events.

In addition, Figs. A.3 and A.4 show the daily closing values of the NASDAQ 100 during the dot-com era and the AI era, respectively, including the trading volume and the trend capturing indicators of the 50-day SMA, a LOWESS. The sharp run-up and subsequent collapse around the 1999–2000 peak are clearly visible. In general, the same trend as in Figs. Figure A.1 and A.2 are observed.

Finally, in Figs. A.6 and A.5, the same illustration is given but the broader market capturing index of SP 500. Beyond the conclusion that the Figs. A.6 and A.5 follow the same trends as the previous indices, its not easy to draw any other conclusion without addition processing of the data. This is because there is a time difference of approximately 100 years between the end of the dot-com bublle and the commence of the AI era.

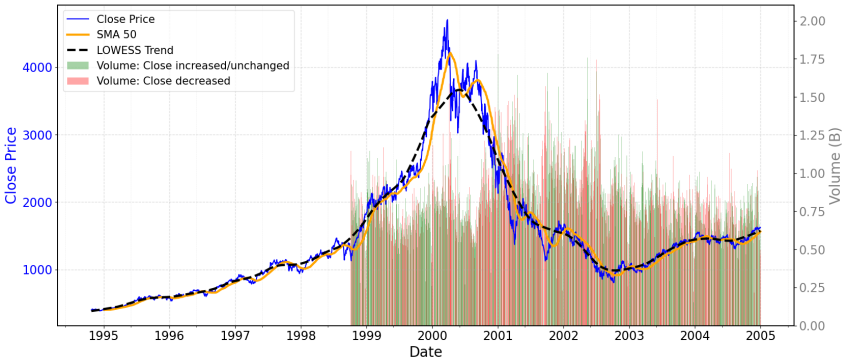


Figure A.3: NASDAQ 100 (Dot-com Era): Close, SMA(50), LOWESS, and volume. Note that, until the moment of writing, the volume data for NASSDAQ 100, TradingView (nd) was available since the second part of 1999.

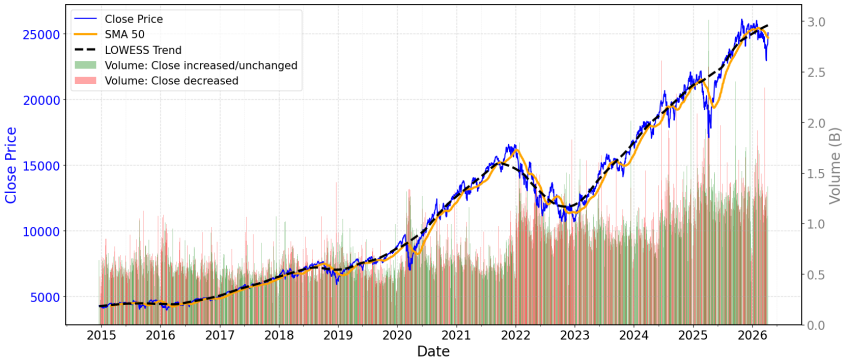


Figure A.4: NASDAQ 100 (AI Era): Close, SMA(50), LOWESS , and volume.

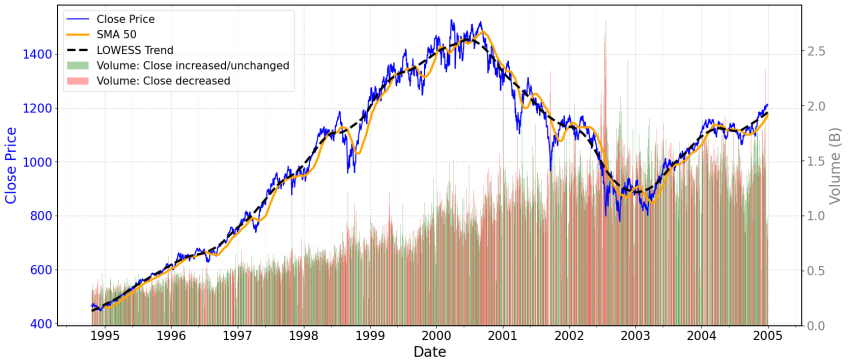


Figure A.5: SP 500 (AI dot-com): Close, SMA(50), LOWESS, and volume.

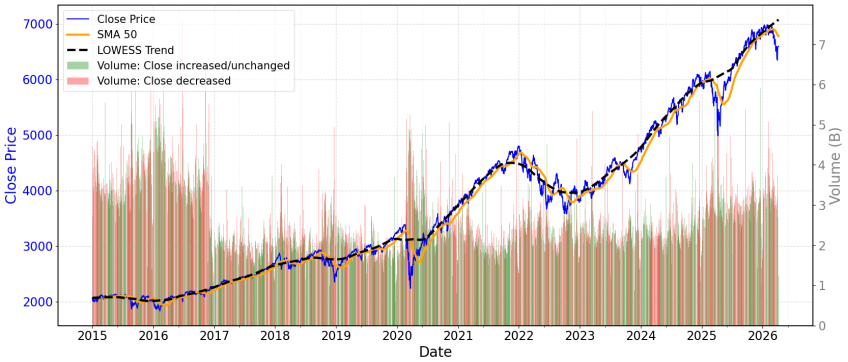


Figure A.6: SP 500 (AI Era): Close, SMA(50), LOWESS , and volume.

A.2 Firm-Level Recovery

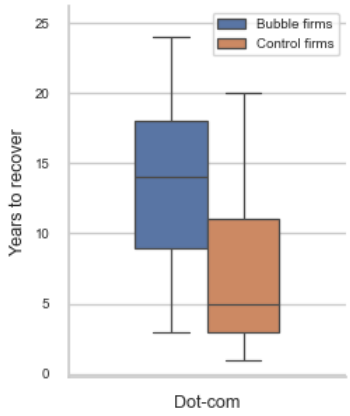


Figure A.7: Distribution of the years that took firms' stock prices to recover after the dot-com bubble.

Figure A.7 shows the distribution of the number of years that dot-com firms required to recover their pre-crash stock-price peak, supporting the recovery analysis discussed in Chapter 4. Firms associated with stronger bubble conditions generally took longer to recover than the control firms.

A.3 Firm-Level Comparability Analysis

The following figures support the firm-level comparability assessment introduced in Chapter 3. They describe how the four comparison groups differ in industry composition and firm maturity, and verify that the firm-level P/S differences reported in Chapter 4 are not merely an artifact of these characteristics. The compared groups are therefore interpreted as benchmark groups rather than perfectly matched populations.

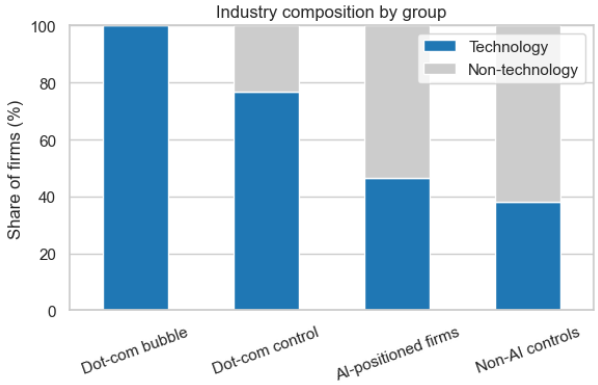


Figure A.8: Industry Composition by Group

Figure A.8 reports the industry composition of each group, expressed as the share of Technology versus non-Technology firms. Within each era, the affected and control groups overlap substantially in industry coverage, whereas the AI era spans a broader and less concentrated set of sectors. This is consistent with the characterization of AI as a general purpose technology adopted across many sectors.

A.3.1 Price-to-Sales Controlled for Industry and Maturity

The next figures repeat the P/S comparison while controlling for industry and firm maturity. The dot-com reference is the peak value, and the AI-era reference uses the 2024, 2025, and 2026 snapshots, consistent with the main analysis. Because the two eras use different sector taxonomies, industry is harmonized into a coarse Technology versus non-Technology classification.

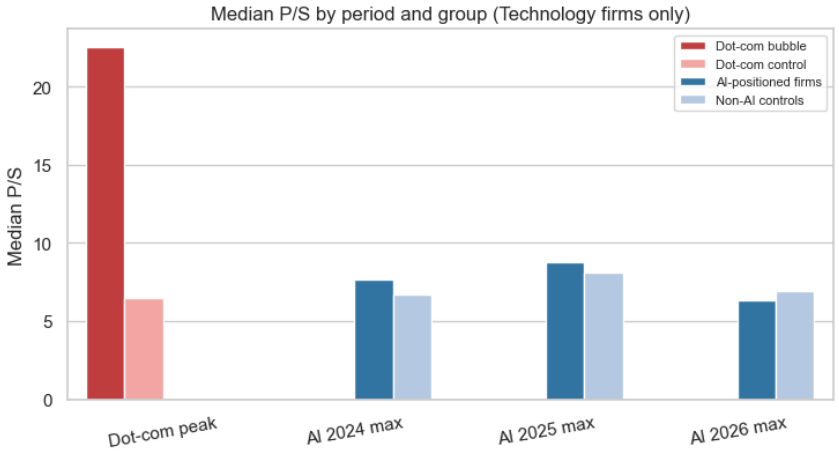


Figure A.9: Median P/S by group for tech firms only

Figure A.9 reports the median P/S by group when the comparison is restricted to Technology firms only, that is, a like-for-like industry comparison. Even within this narrower set, the dot-com bubble group remains clearly above the AI-era groups, while the AI-positioned and non-AI control groups remain close to each other.

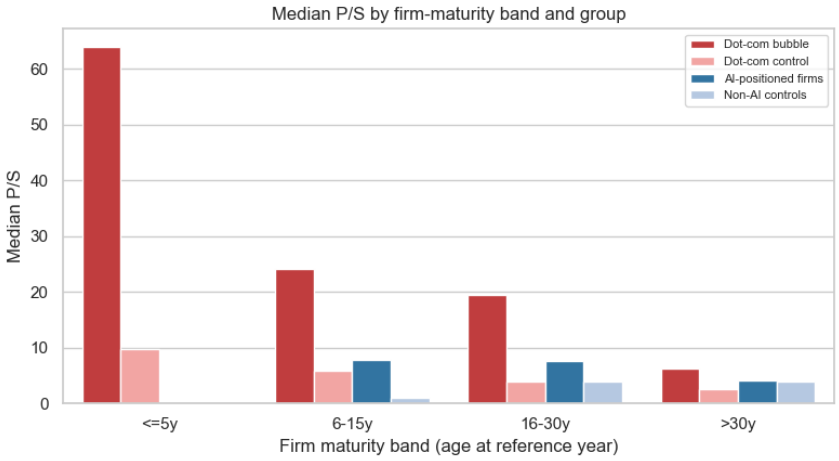


Figure A.10: Median P/S grouped by firm maturity

Figure A.10 reports the median P/S within firm-maturity bands defined by firm age at the reference year. The comparison shows that the differences between groups persist within comparable maturity bands, indicating that the results are not driven only by differences in firm age.

Figure A.11 extends the previous figure by showing the median P/S for each maturity band across the valuation periods (the dot-com peak and the AI 2024–2026 snapshots). This jointly captures the maturity and the time dimensions, and confirms that the lower P/S levels of the AI era relative to the dot-com peak hold across maturity bands and across the observed years.

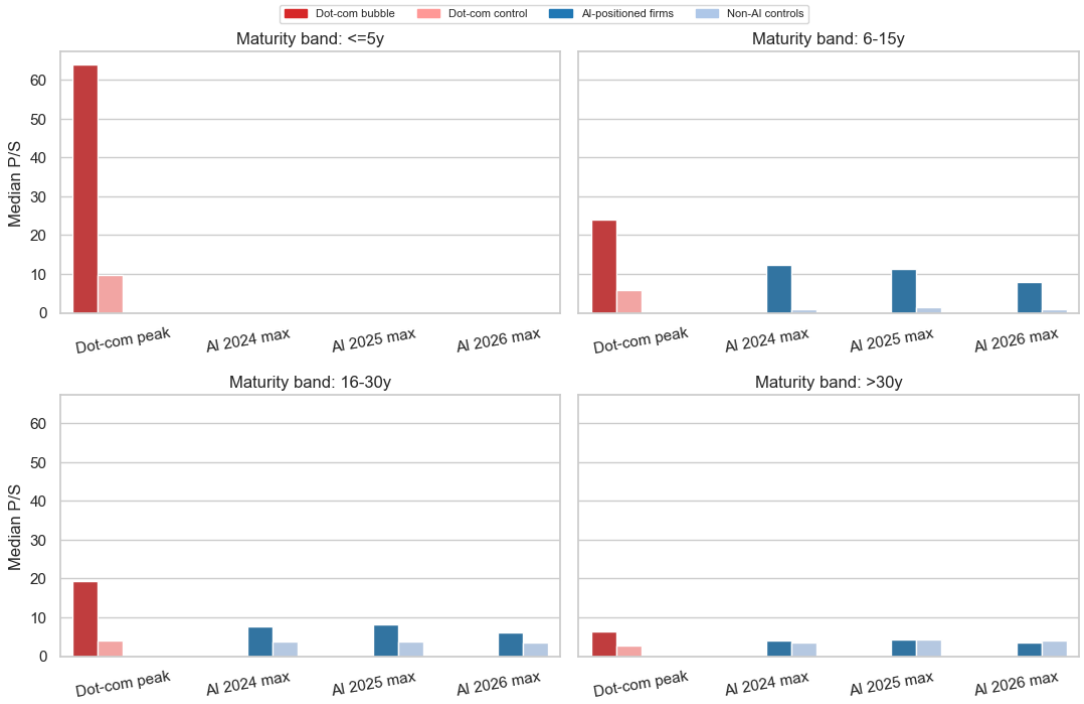


Figure A.11: Median P/S grouped by firm maturity and valuation years

B. Appendix: Tables

This appendix collects the supporting tables referenced in the main text. It first summarizes the closely related prior studies, and then reports the firm-level descriptive tables that complement Chapter 4.

B.1 Previous Research Comparison Table

Table B.1 summarizes the most closely related prior studies that compare the AI and dot-com periods. For each study, it reports the type of data, the analytical approach, the main gap it leaves, and its overall conclusion. The table expands the discussion of the research gap in Section 2.5 and highlights that existing work rarely combines market-level cycle similarity with firm-level evidence on valuation and profitability.

Table B.1: Summary of previous Literature on AI vs. Dot-com Bubbles

	Study	Types	Analysis	Gap	Conclusions
1	Dobre et al. (2020)	Index analysis: Aggregated indices from 228 dot-com firms vs 426 AI firms	Correlation + stands for Autoregressive Distributed Lag (ARDL) regression	Period ends pre-LLM (2018). No firm-level fundamentals (profits, valuation)	AI market shows patterns similar to dot-com bubble
2	Potrykus (2024)	2 Indices analysis + 41 AI firm signals	DTW + Generalized Supremum Augmented Dickey-Fuller (GSADF) bubble test	Focus on price pattern similarity only. Limited firm-level financial analysis with no firms valuation metrics	Strong resemblance in price trajectories → AI follows similar boom dynamics to dot-com
3	Defend et al. (2026)Defend et al. (2026)	Portfolio-level market analysis	Factor exposure, P/E, Capital Expenditure (CapEx), autocorrelation	Portfolio aggregation → hides firm heterogeneity. No firm-level comparison. No direct DTW-style historical pattern matching	Detects speculative bubbles in AI-related stocks, but bubbles are less widespread than dot-com
4	Subrahmanyam and Beainy (2026)	Firm-level analysis, with only Large-cap firms (3 from dot-com vs 3 from AI)	Financial ratios of metrics: Revenue/CapEx, Free Cash Flow (FCF) margins, profitability	Focus on large-cap firms only (sample bias). No full market-wide or index-level dynamics.	AI is not a bubble, and has strong fundamentals. Growth is economically justified.
5	Chelikavada and Bennett (2026)	research activity analysis + Indices Analysis: S&P 500, NASDAQ, New York Stock Exchange (NYSE), Volatility IndeX (VIX)	System of National Accounts (SNA) + Graph Convolutional Network (GCN) + Long Short-Term Memory (LSTM) + Generalized Autoregressive Conditional Heteroskedasticity (GARCH)/Autoregressive model with eXogenous variables (AR-X) Focus on: citation networks + scientific hype signals	No firm-level analysis. Basic indices analysis (only hype signals). Inconclusive results.	Dot-com patterns are different from AI patterns → Either: AI is NOT a bubble OR a new type of bubble.
6	Cao (2025)	theoretical model of tech booms	Two-sector Dynamic Stochastic General Equilibrium (DSGE) macroeconomic model	Purely theoretical model-driven research. No direct statistical validation	AI can conditionally behave like dot-com depending on spillover speed & capital supply elasticity
7	Floridi (2024)	Conceptual analysis. dot-com, crypto, telecom vs AI	bubble characteristics (euphoria, valuation gaps, hype) with focus on ethics, regulation, investor behavior	fully conceptual without measure or validation or any empirical data analysis.	AI hype strongly resembles past tech bubbles

B.2 Chapter 4 Tables

The following tables report the mean and median values of the operating-performance measures across the dot-com and AI eras. They complement the firm-level analysis in Chapter 4 and are reported here

because the main text focuses on the median, which is more robust to the extreme outliers present in both samples.

Table B.2 reports the EV/EBITDA multiple, which relates enterprise value to operating earnings. The large gap between the mean and median, especially for the dot-com bubble firms, reflects strong right-skewness driven by a small number of extreme observations. In median terms, the AI-era and dot-com groups are relatively close, indicating that the difference narrows once valuation is compared with operating earnings.

Table B.2: Mean and median EV/EBITDA multiples across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms)

	Affected firms		Control firms	
Group	Mean	Median	Mean	Median
Dot-com era	590.92	19.23	34.29	26.82
AI era	20.25	16.91	20.09	17.74

Table B.3 reports the EBITDA margin, which measures operating profitability relative to revenue. The negative means for the dot-com bubble firms and the AI control firms again reflect a few firms with large negative margins, while the medians indicate that the AI-era firms are generally more profitable than the dot-com firms.

Table B.3: Mean and median EBITDA margins across dot-com and AI firm groups (Affected firms are the dot-com bubble firms or AI-positioned firms)

	Affected firms		Control firms	
Group	Mean	Median	Mean	Median
Dot-com era	-148.84	9.86	15.24	13.69
AI era	24.83	26.10	-17.09	22.95



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